Undergraduate
Program Course Curriculum



Department of Materials Engineering

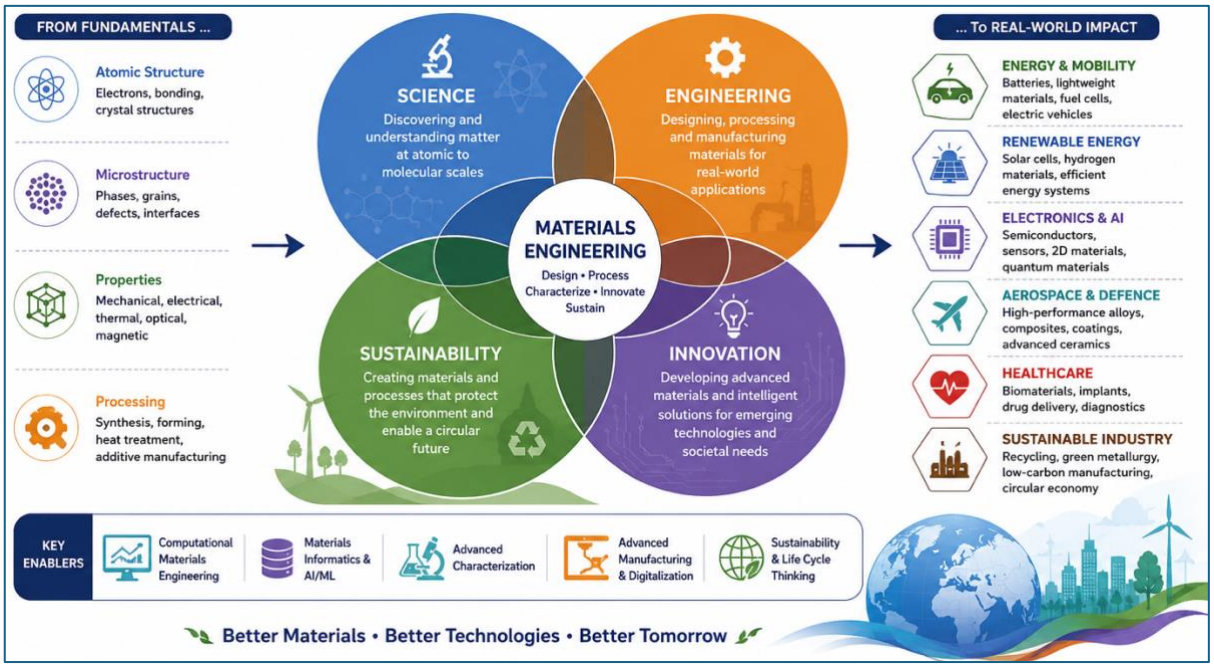
Indian Institute of Technology Jodhpur

Department of Metallurgical and Materials Engineering

Curriculum of UG programme in Materials Engineering

1. Introduction

Materials are at the heart of modern technological advancements and play a critical role in addressing global challenges related to energy, sustainability, healthcare, transportation, infrastructure, electronics, aerospace, defence, and advanced manufacturing. The development of next-generation technologies such as electric vehicles, renewable energy systems, semiconductors, smart devices, hydrogen technologies, and additive manufacturing critically depends on the discovery, design, processing, and deployment of advanced materials with tailored properties and enhanced performance.



Globally, Materials Engineering has emerged as a strategic and interdisciplinary domain that integrates principles of physics, chemistry, mathematics, biology, computation, and engineering to understand and control the relationships among processing, structure, properties, and performance of materials. Materials innovation is central to Industry 4.0, sustainable manufacturing, energy transition, and national technology missions focused on semiconductor manufacturing, critical materials, and green technologies.

To address these emerging technological and societal needs, the Department of Metallurgical and Materials Engineering at IIT Jodhpur proposes a comprehensive B.Tech. programme in Materials Engineering aimed at imparting high-quality education and training in both traditional and emerging areas of materials science and engineering. The programme is designed to equip students with strong foundations in materials design, processing, characterization, computational modelling, manufacturing, and process metallurgy, along with exposure to modern data-driven and AI-assisted materials engineering approaches.

The proposed curriculum emphasizes the fundamental understanding of the processing–structure–properties–performance paradigm of materials and integrates experimental, computational, and analytical methodologies for solving real-world engineering

challenges. The programme encompasses four major thematic streams: Structural Materials Functional Materials Computational Materials Engineering Process Metallurgy

The curriculum includes foundational courses, programme core courses, laboratory training, interdisciplinary electives, and specialization pathways that together provide both breadth and depth in Materials Engineering. Special emphasis is placed on emerging domains such as energy materials, electronic and semiconductor materials, advanced structural alloys, additive manufacturing, materials informatics, computational materials engineering, and sustainable metallurgy.

The programme also aims to develop industry-ready graduates with strong experimental, computational, analytical, and problem-solving skills relevant to academia, research laboratories, advanced manufacturing industries, energy technologies, semiconductor industries, and high-technology sectors. In addition, the curriculum encourages innovation, interdisciplinary learning, ethical engineering practice, entrepreneurship, and lifelong learning.

To further enable focused learning and specialization, students will have the opportunity to pursue departmental specialization tracks in one of the four major streams through additional advanced coursework and project-based learning. This structure is expected to provide students with the flexibility to align their academic training with their interests, emerging industrial demands, and future research opportunities.

Overall, the proposed B.Tech. programme in Materials Engineering is envisioned to create highly skilled engineers and innovators capable of contributing to the development of advanced materials and sustainable technologies that are essential for the future needs of the nation and society.

2. Objectives of the Program

I. To provide strong foundational knowledge in materials engineering encompassing structure-processing-properties-performance relationships of materials.
II. To develop experimental, computational, analytical, and problem-solving skills for the design, characterization, processing, and performance evaluation of materials.
III. To train students in contemporary and evolving areas such as energy materials, electronic materials, computational materials engineering, sustainable metallurgy, and materials informatics.
IV. To prepare graduates capable of addressing industrial, technological, and societal challenges through interdisciplinary and innovation-driven approaches.
V. To inculcate professional ethics, lifelong learning, leadership, teamwork, and effective communication skills for successful professional careers.
VI. To provide opportunities for research-oriented learning in key domains of Materials Engineering.

3. Expected Graduate Attributes

Graduates of the B.Tech. programme in Materials Engineering are expected to possess:

1. Strong foundational knowledge in the science and engineering of metallic, ceramic, polymeric, electronic, and composite materials.

2. Understanding of the processing–structure–properties–performance relationships governing materials behaviour and engineering applications.

3. Ability to integrate experimental, computational, analytical, and data-driven approaches for solving materials engineering problems.

4. Knowledge of modern materials technologies relevant to energy, electronics, sustainability, manufacturing, healthcare, aerospace, and advanced engineering applications.

5. Capability to work effectively in interdisciplinary environments with professional ethics, social responsibility, leadership, and communication skills.

6. Adaptability towards emerging technologies, innovation-driven research, entrepreneurship, and lifelong learning.

4. Learning Outcomes

Upon successful completion of the programme, students will be able to:

I. Apply principles of thermodynamics, kinetics, mechanics, and materials science for understanding and designing engineering materials and processes.

II. Correlate processing, structure, properties, and performance of materials across different length scales and application domains.

III. Use experimental techniques for synthesis, processing, characterization, testing, and performance evaluation of materials.

IV. Apply computational, simulation, and data-driven tools for modelling, prediction, and optimization of materials behaviour and processes.

V. Analyse and design materials systems for applications related to energy storage, electronics, structural engineering, manufacturing, sustainability, and advanced technologies.

VI. Identify, formulate, and solve complex materials engineering problems individually and collaboratively using scientific and engineering principles.

VII. Communicate technical concepts effectively and demonstrate professional ethics, teamwork, and awareness of societal, environmental, and sustainability considerations in engineering practice.

5. Course categories and credit distribution in the B.Tech. programme

Table 1

S.N.	Course Type	Course Category	Credits	Minimum Total
1	Institute Core (I)	Engineering (IE)	16	39 (27.6%)
		Science (IS)	17	
		HSS (IH)	06	
2	Programme Core (P)	Programme Compulsory (PC)	66	**90 (63.8%)**
		Programme Electives (PE)	18	
		(B.Tech.) Project (PP)	6	
3	Open (O)	Electives (OE)	12	12 (8.5%)
			Total (Graded)	**141 (100%)**
4	Essential Audit (Non-taught)	Humanities (NH)	3	**12**
		Engineering (NE)	3	
		Industry-Academia Summer Internship outside IIT Jodhpur (two summers, minimum 45 days)	4	
		Design Credit	2	
			Total for award of Degree (Graded + Non-Graded)	**153**

List of Institute-wide courses:

Institute-wide Engineering (IE) Courses (16 credits):

1. Engineering Mechanics [3-0-0]
2. Introduction to Computer Science [3-0-2]
3. Introduction to Machine Learning [3-0-2]
4. Engineering Workshop [0-0-2]
5. Engineering Drawing [0-0-2]
6. Environment and Sustainability [3-0-0]

Institute-wide Science (IS) courses (17 credits):

1. Physics [3-0-0]
2. Physics Lab [0-0-2]
3. Chemistry [3-0-0]
4. Chemistry Lab [0-0-2]
5. Mathematics I [2-1-0]
6. Mathematics II [2-1-0]
7. Introduction to Bioengineering [2-0-2]

Institute-wide NH courses (3 credits):

1. Communication Skill [0-0-2]
2. Introduction to Professional Ethics [1-0-0]
3. Performing Arts /Sports [0-0-1]
4. Social Connect and Responsibilities [0-0-1]

Institute-wide NE courses (3 credits):

1. Engineering Design I [0-0-2]
2. Engineering Design II [0-0-2]
3. Introduction to Profession [0-0-2]

Note: Physics and Chemistry courses will be offered in both semesters. Other institute-wide courses will be offered once a year, as per the semester-wise plan as mentioned in Table 3.

Table 2: Credit Requirement Table

S.N.	Course Type	Course Category	B.Tech. Regular	Dual Degree (B.Tech +PhD)	B.Tech. With Minor
1.	Institute Core (I)	Engineering (IE)	16	16	16
		Science (IS)	17	17	17
		Humanities (IH)	6	6	6
2.	Programme Core (P)	Programme Compulsory (PC)	66	66	66
		Programme Electives (PE)	18	18	18
		B.Tech. Project (PP)	6	6	6
3.	Open (O)	Electives (OE)	12	12	0
4.	Additional Credits	Programme Compulsory (PC) (Fixed Courses Prescribed by the offering Academic Unit.)	0	0	18
5.	Ph.D.	Compulsory (DC)	0	12	0
		Electives (DE)	0	12	0
		PhD. Thesis (MP)	0	64	0
		Total Graded	**141**	**229**	**147**
6.	Essential Audit (Non-taught)	Humanities (NH)	3	3	3
		Engineering (NE)	3	3	3
		Design/Practical Experience (ND)	2	2	2
		Industry-Academia Summer Internship outside IIT Jodhpur (two summers, minimum 45 days)	4	4	4
		Total Graded + Essential Audit	**153**	**241**	**159**

Table 3: Semester-wise plan for B.Tech. in Materials Engineering

Cat	Course	LTP	CH	NC	GC	Cat	Course	LTP	CH	NC	GC
	I Semester						**II Semester**				
IE	Engineering Mechanics	3-0-0	3	-	3	IE	Introduction to Machine Learning	3-0-2	5	-	4
IE	Introduction to Computer Science	3-0-2	5	-	4	IS	Chemistry	3-0-0	3	-	3
IS	Physics	3-0-0	3	-	3	IS	Introduction to Bioengineering	2-0-2	4	-	3
IH	HSS I	3-0-0	3		3	IS	Chemistry Lab	0-0-2	2	-	1
IS	Physics Lab	0-0-2	2	-	1						
IS	Mathematics I	2-1-0	3	-	3	IS	Mathematics II	2-1-0	3	-	3
IE	Engineering Drawing	0-0-2	2	-	1	IE	Engineering Workshop	0-0-2	1	-	1
NE	Engineering Design I	0-0-2	2	1	-	NE	Engineering Design II	0-0-2	2	1	-
NH	Communication Skill	0-0-2	2	1	-	IH	HSS II	3-0-0	3		3
NH	Social Connect and responsibilities	0-0-1	1	0.5	-	NE	Introduction to Profession	0-0-2	2	1	
NH	Performing Arts /Sports	0-0-1	1	0.5	-						-
Total					**18**	**Total**					**18**
	III Semester						**IV Semester**				
PC	Numerical Methods for Materials Engineering	3-0-0	3	-	3	PC	Kinetics of Materials	3-0-0	3	-	3
PC	Materials Thermodynamics	3-0-0	3	-	3	PC	Materials for Functional Applications	3-0-0	3	-	3
PC	Materials Science and Engineering	3-0-0	3	-	3	PC	Materials for Energy Conversion and Storage	3-0-0	3	-	3
PC	Mechanics of Solids	3-1-0	4	-	4	PC	Transport Phenomena in Materials	3-0-0	3		3
IE	Environment and Sustainability	3-0-0	3		3	PC	Mechanical Behaviour of Materials	3-0-0	3		3
PC	Numerical Methods and Materials Engineering Lab	0-0-4	4		2	PC	Materials for Functional Applications Lab	0-0-4	4		2
Total					**18**	**Total**				**-**	**17**
	Summer Semester Internship (Minimum 45 days, Outside IIT Jodhpur)										
	V Semester						**VI Semester**				
PC	Materials Degradation	3-0-0	3		3	PC	Metallurgy of Joining	3-0-0	3	-	3
PC	Phase Transformations	3-0-0	3		3	PC	Materials Characterization	3-0-0	3	-	3
PC	Materials Processing and Manufacturing	3-0-0	3	-	3	PC	Principles of Metal Extraction	3-0-0	3	-	3
PC	Synthesis and Processing of Semiconductor materials	3-0-0	3		3	PC	Processing of polymers and ceramics	3-0-0	3	-	3

PC	Physical Metallurgy and Materials Modelling Lab	0-0-4	4		2	PC	Materials Processing and Characterization Lab	0-0-4	4	-	2
PC	Computational Modelling of Materials	3-0-0	3	-	3	PC	Machine Learning for Materials Informatics	3-0-0	3	-	3
OE	Open Elective	3-0-0	3	-	3	PE	Program Elective	3-0-0	3		3
NH	Professional Ethics	1-0-0	1	1	-						
Total		**19-0-4**	**21**	**1**	**20**	**Total**		**17-0-6**	**23**		**20**
	Summer Semester Internship (Minimum 45 days, Outside IIT Jodhpur)										
	VII Semester						**VIII Semester**				
PP	B.Tech. Project 1	0-0-3			3	PP	B.Tech. Project 2	0-0-3			3
PE/ OE	Programme/Open Electives				12	PE/ OE	Programme/Open Electives				12
Total					**15**	**Total**				**-**	**15**
Total of Graded and Non-Graded Credit										6	**141**
Industry-Academia Summer Internship										4	
Non-Graded Design Credits										2	
Grand Total										12	**153**

Table 4: List of core courses for Minor in Materials Engineering

Course	L-T-P	CH	NC	GC
Introduction to Materials Science & Engineering	3-0-0	3	-	3
Introduction to Functional Materials	3-0-0	3	-	3
Materials Processing and Manufacturing	3-0-0	3	-	3
Materials Thermodynamics	3-0-0	3	-	3
Computational Modelling of Materials	3-0-0	3	-	3
Materials Characterization	3-0-0	3	-	3
Total				18

6. List of Programme Compulsory Courses

Table 5 : Programme Compulsory Courses for B.Tech. in Materials Engineering

Sr. No.	Course Name	L-T-P	CH	GC	Page No.
1	Materials Science and Engineering	3-0-0	3	3	12
2	Materials Thermodynamics	3-0-0	3	3	13
3	Numerical Methods for Materials Engineering	3-0-0	3	3	14
4	Mechanics of Solids	3-1-0	4	4	15
5	Numerical Methods and Materials Engineering Lab	0-0-4	4	2	16
6	Kinetics of Materials	3-0-0	3	3	18
7	Materials for Functional Applications	3-0-0	3	3	19
8	Materials for Energy Conversion and Storage	3-0-0	3	3	20
9	Transport Phenomena in Materials	3-0-0	3	3	21
10	Mechanical Behaviour of Materials	3-0-0	3	3	22
11	Materials for Functional Applications Lab	0-0-4	4	2	23
12	Materials Degradation	3-0-0	3	3	24
13	Phase Transformations	3-0-0	3	3	25
14	Materials Processing and Manufacturing	3-0-0	3	3	26
15	Synthesis and Processing of Semiconductor materials	3-0-0	3	3	27
16	Computational Modelling of Materials	3-0-0	3	3	28
17	Physical Metallurgy and Materials Modelling Lab	0-0-4	4	2	29
18	Metallurgy of Joining	3-0-0	3	3	30
19	Materials Characterization	3-0-0	3	3	31
20	Principles of Metal Extraction	3-0-0	3	3	32
21	Machine Learning for Materials Informatics	3-0-0	3	3	33
22	Processing of polymers and ceramics	3-0-0	3	3	34
23	Materials Processing and Characterization Lab	0-0-4	4	2	35
	Total			66	

7. List of Stream-wise Programme Elective Courses

Table 6: Stream-wise Programme Electives Courses for B.Tech. in Materials Engineering

Stream	Sr. No.	Course Name	Level	L-T-P	CH	GC	Page No.
Structural Materials	1	Processing-Microstructure and Properties of Steels	400	3-0-0	3	3	37
	2	Failure of Materials and Prevention	400	3-0-0	3	3	38
	3	Composites	400	3-0-0	3	3	39
	4	Heat-Treatment of Metals and Alloys	400	3-0-0	3	3	40
	5	Solidification of Metals and Alloys	700	3-0-0	3	3	41
	6	Light Metals & alloys	700	3-0-0	3	3	42
	7	High Temperature Materials	700	3-0-0	3	3	43
Functional Materials	8	Battery Materials	400	3-0-0	3	3	44
	9	Functional Ceramics	400	3-0-0	3	3	45
	10	Magnetism and Magnetic Materials	400	3-0-0	3	3	46
	11	Solar Cell Materials and Devices	700	3-0-0	3	3	47
	12	Advanced Semiconductor Materials and Device Applications	700	2-0-2	3	3	48
	13	Functional 2D Materials	700	2-0-2	3	3	49
	14	Functional Materials	700	3-0-0	3	3	51
Computational Materials	15	Computational Thermodynamics	400	3-0-0	3	3	52
	16	Introduction to Dislocation Dynamics	400	3-0-0	3	3	53
	17	Machine Learning for Materials Design	700	2-0-2	3	3	54
	18	First-Principles Calculations	700	2-0-2	3	3	55
	19	Advanced Molecular Dynamics	700	3-0-0	3	3	56
	20	ICME	700	3-0-0	3	3	57
Process Metallurgy	21	Mineral Beneficiation	400	3-0-0	3	3	59
	22	Extraction of Nonferrous Metals	400	3-0-0	3	3	60
	23	Electrometallurgy	400	3-0-0	3	3	61
	24	Iron and Steel Making	400	3-0-0	3	3	62

Detailed Course Content of Programme
Compulsory Courses

8. Detailed Course Content of Programme Compulsory Courses

Title	**Materials Science and Engineering**	Number	MTL2XX0
Department	Materials Engineering	L–T–P [C]	3–0–0 [3]
Offered for	B.Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Objectives

1. Provide a basis understanding of material science.
2. Introduce students to the relationship between structure, properties, processing, and performance of materials, for building a foundation for advance materials study.

Learning Outcomes

1- Explain the fundamental concepts of materials science, including the relationship between structure, properties, processing, and performance.

Contents

Fundamental of material science [3 lectures]: Introduction to Materials Science & Engineering, Classification of materials (metals, ceramics, polymers, composites), Structure–Property–Processing–Performance relationship, Overview of advanced materials.

Atomic structure, Bonding & Crystal structure [6 lectures]: Atomic structure and electronic configuration, Bonding force and energies, Ionic bonding, Covalent Bonding, Metallic Bonding. Crystal systems and Bravais lattices, Miller indices (planes and directions), X-ray diffraction and Bragg's law

Imperfections in Solids [5 lectures]: Point defects, Line defects and dislocations, Interfacial defects, Bulk or volume defects, significance of defects in materials. Grain boundaries.

Mechanical Properties of Materials [5 lectures]: Concept of stress and Strain, Elastic deformation, Plastic deformation, Shear, Hardness

Electrical properties of materials [5 lectures]: Electrical conduction in solids, Drift velocity, Energy bands: valence band and conduction band, Intrinsic semiconductors, Extrinsic semiconductors. Diodes, Transistors, Solar cells, Ferroelectricity, Piezoelectricity, Solar cell.

Magnetic properties of materials [5 lectures]: Magnetic dipole moment, Electron spin, Diamagnetism, Paramagnetic, Ferromagnetism, Antiferromagnetism, Ferrimagnetism, Domain structure, Hysteresis loop, soft vs hard magnetic materials.

Optical properties of materials [5 lectures]: Reflection, refraction, absorption, transmission, Refractive index, Band gap and color of materials, Light interaction with solids, Optical fibers.

Thermal properties of materials [5 lectures]: Basic laws of thermodynamics, specific heat, entropy, thermal conductivity, thermal stresses.

Text Books

1. W. D. Callister Jr. and D. G. Rethwisch, Materials Science and Engineering: An Introduction, 10th ed. Hoboken, NJ, USA: John Wiley & Sons, 2020.

References

1. Raghavan, V. (2004). Materials science and engineering: A first course. Prentice Hall India.

Online Course Material

Massachusetts Institute of Technology. (2005). Fundamentals of Materials Science (3.012). MIT OpenCourseWare. https://ocw.mit.edu/courses/3-012-fundamentals-of-materials-science-fall-2005/

Title	**Materials Thermodynamics**	Number	MTL 2XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Objectives

1. Provide a basis for describing and understanding the stability of various forms of matter including metals, polymers, and multi-phase systems.
2. Provide a basis for predicting the properties and phase behavior of equilibrated systems as a function of their chemical content and interfacial characteristics.

Learning Outcomes

1. Establish the conditions for stability of a material and derive its properties.
2. Control the evolution of microstructures with respect to different thermal and compositional parameters.

Contents

Introduction to thermodynamics of materials [2 lectures]: Introduction – systems and boundaries, concept of state, state and path functions, equilibrium definition, thermodynamics variables – extensive and intensive.

Laws of thermodynamics [10 lectures]: First law of thermodynamics, internal energy, heat capacity, adiabatic processes, second law of thermodynamics, entropy and statistical interpretation of entropy, third law of thermodynamics and thermodynamic functions: Helmholtz free energy, enthalpy, Gibbs free energy, chemical potential, Maxwell's relations, Gibbs-Helmholtz relation.

Single component systems [2 lectures]: Phase equilibrium in single component systems, Gibbs free energy variation with pressure and temperature, P-T diagrams, Clausius-Clapeyron equation.

Behaviour of gases and solution thermodynamics [8 lectures]: Ideal and non-ideal gases, activity, fugacity, partial molar quantities, Raoult's law and Henry's law, Gibbs-Duhem equation, Gibbs free energy of formation of a solution (ideal, non-ideal), regular solution model, statistical model of solutions, Flory-Huggins theory and Flory interaction parameter (χ), entropy of mixing for polymeric chains.

Composition and binary phase diagrams [8 lectures]: Thermodynamic activity, criterion for phase stability, Gibbs free energy composition diagrams, activity-composition diagrams, application of regular solution model to phase diagrams, Gibbs phase rule, phase behavior in polymer systems: LCST/UCST, phase separation in polymer blends.

Surface and Interfacial Thermodynamics [4 lectures]: Adsorption and interphase: work of adhesion, thermodynamics of fiber-matrix interfaces.

Reactions involving gases and condensed phase [5 lectures]: Reaction equilibrium a gas mixture and the equilibrium constant, effect of pressure and temperature on the equilibrium constant, standard Gibbs free energy variation with temperature, Ellingham diagrams, effect of phase transformations.

Text Books

1. Gaskell, D.R. and Laughlin, D.E., (2018), Introduction to Thermodynamics of Materials, 6th Edition CRC Press

References

1. Hillert, M., (2008), Phase Equilibria, Phase Diagrams and Phase Transformations Their Thermodynamic Basis, Second edition, Cambridge University Press
2. Dehoff, R., (2006), Thermodynamics in Materials Science, 2nd Edition, CRC Press

Online Course Material

1. Ceder, G., and Anton Van der Ven, A., 3.20 Materials at Equilibrium (SMA 5111). Fall 2003. Massachusetts Institute of Technology: MIT OpenCourseWare, https://ocw.mit.edu. License: Creative Commons BY-NCSA.
2. Thermodynamics and kinetics of materials: Dr. Saswata Bhattacharya, Material Science and Metallurgical Engineering, IIT Hyderabad. https://nptel.ac.in/courses/113106109

Title	**Numerical Methods for Materials Engineering**	Number	MTL 2XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Objectives:

1. Introduce numerical methods and scientific-computing tools for solving materials-engineering problems.
2. Enable students to analyze experimental and simulation data quantitatively and automate common calculations.

Learning outcomes:

1. Read, process and visualize typical materials datasets.
2. Apply basic numerical methods to materials problems.
3. Implement small, reusable scripts for routine calculations and analyses.

Contents

*Materials Data handling and mathematical foundations (7 lectures):*Data handling and exploratory analysis - Reading and organizing materials datasets, units, metadata. Descriptive statistics for experimental data. Basic probability and statistics for engineering data. Vectors and matrices as data containers. Matrix operations. Power series and Taylor series as local approximations to functions. Brief discussion of numerical error: truncation vs round-off. Fourier Transform and application on diffraction and FTIR spectroscopy

Curve fitting, regression, interpolation (9 lectures): Linear regression in one variable; analytical formula and matrix form, General linear least squares, Polynomial regression, Overfitting vs underfitting, interpolation vs regression, Polynomial interpolation, Runge phenomenon, Splines and piecewise interpolation, Brief idea of Fourier, Nonlinear regression and model fitting on stress-strain, saturation growth, diffusion

Numerical differentiation and integration (9 lectures): Finite-difference approximations: forward, backward, central differences; order of accuracy. Differentiating discrete experimental data: slopes of stress–strain curves, dD/dT for diffusion data, Sensitivity to noise; smoothing vs differentiation trade-offs. Rectangle rule, trapezoidal rule, Simpson's 1/3 rule; error orders. Composite rules, non-uniform grids; integrating experimental data with uneven spacing.

Linear systems and eigenvalue (9 lectures): Solving linear systems, Formulating engineering problems as Ax=b: mixture rules, force balance, simple thermal networks. Interpreting least-squares regression as solver. Rank deficiency and ill-conditioning. Basic idea of eigenvalues and eigenvectors, modes and stability. Concept of eigenvalues and eigenvectors relevant to diffusion and vibration.

Initial-value ODEs and simple diffusion/thermal problems (5 lectures): ODEs – fundamentals and explicit methods, Initial-value problems with physical examples of cooling of a body, and simple kinetics. Euler's method and improved Euler. Runge–Kutta methods. Diffusion/heat-conduction in 1D.

Textbooks

1. S. C. Chapra, Applied Numerical Methods with MATLAB for Engineers and Scientists, 5th Edition, McGraw-Hill, 2022.
2. M. L. Boas, Mathematical Methods in the Physical Sciences, 3rd Edition, Wiley, 2006.

References

1. E. Kreyszig, Advanced Engineering Mathematics, 10th Edition 2010, John Wiley & Sons Inc.
2. S. S. Sastry, Introductory Methods of Numerical Analysis, 5th Edition, PHI Learning. 2012.
3. K. F. Riley, M. P. Hobson, and S. J. Bence, Mathematical Methods for Physics and Engineering: A Comprehensive Guide, 3rd Edition, Cambridge University Press, 2006.

Online Course Materials

1. Lal, R. and Banerjee S., Numerical Analysis, NPTEL Course Material, Department of Mathematics, IIT Roorkee, https://nptel.ac.in/courses/111107062/
2. Usha, R., Numerical Analysis, NPTEL Course Material, Department of Mathematics, IIT Madras, https://nptel.ac.in/courses/111106101/

Title	**Mechanics of Solids**	Number	MTL 2XX0
Department	Materials Engineering	L – T – P [C]	3 – 1 – 0 [4]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Objectives

1. To inculcate fundamental concepts of stress, strain, and deformation of solids
2. Fundamentals of applying equilibrium, compatibility, and forced deformation relationships to structural elements

Learning Outcomes

The students are expected to have the ability to:

1. Apply the fundamentals to analyze structural members and gain physical insight into their stress and strain distributions.
2. Ability to design structural members using dimensions, force-displacement relationships, boundary conditions, loading, and allowable stress

Contents

Concept of stress [7 Lectures]: Surface and body forces, concept of stress at a point, stress equilibrium, plane stress case: transformation of stresses at a point, principal stresses, and Mohr's circle, St. Venant's principles.
Concept of strain [5 Lectures]: Displacement field, strain at a point, compatibility relations, volumetric strain, plane strain case: transformation of strain at a point, principal strains and Mohr's circle, Strain Rosette;
Material behaviour and mathematical model [4 Lectures]: Concepts of elasticity, response of real materials, Generalized Hooke's law (without and with thermal strains) for isotropic materials, orthotropic, and anisotropic materials.
Application of Elasticity:
One-dimensional axial deformation [2 Lectures]: axial force diagram, basic relations for axial deformations, Stress concentration.
Torsion of circular members [3 Lectures]: Basic relations for torsion of circular members, stress distribution of torsion members.
Pressure Vessels [3 Lectures]: Thin-walled pressure vessel theory, thin-walled spheres, thin-walled cylinders.
Beams [7 Lectures]: Shear force and moment diagrams; moment-curvature relationship for pure bending of beams with symmetric cross-section; bending stress, shear stress; cases of combined stresses; deflection due to bending.
Buckling [3 Lectures]: Concept of elastic instability, Introduction to column buckling, Euler's formula.
Energy methods [5 Lectures]: Concept of strain energy; strain energy for simple structural elements; Castigliano's theorems; principle of minimum potential energy.

Text/Reference books

1. Crandall, S.H., Dahl, N.C., and Lardner, T. J., (1978), An Introduction to the Mechanics of Solids, McGraw Hill, Second Ed. with SI Units
2. Popov, E.P., (1990), Engineering Mechanics of Solids, First Ed. Prentice Hall
3. Beer, F.P, Johnston, E.R. and De Wolf, J.T., (2004), Mechanics of Materials, Tata McGrawHill.
4. Meriam, J.L. and Kraige, L.G., (1980), Engineering Mechanics, Vol. 1: Statics, 2nd Ed. John Wiley

Online course material

Prof. Ajeet Kumar, NPTEL Course, Solid Mechanics, IIT Delhi https://nptel.ac.in/courses/112102284
Prof. Priyanka Ghosh, NPTEL Course, Mechanics of Solids, IIT Kanpur
https://nptel.ac.in/courses/10510416

Title	**Numerical Methods and Materials Engineering Lab**	Number	MTL 2XX0
Department	Materials Engineering	L – T – P [C]	0 – 0 – 4 [2]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Objectives

1. Connect small-scale atomic structures to large-scale material properties through hands-on testing.
2. Using lab equipment and generating raw datasets to draw accurate scientific conclusions.
3. Translate physical material behaviours into simpler and solvable mathematical models..

Learning outcomes

1. Understand how a material's properties and its processing route change its internal structure and physical strength.
2. Extract important material properties from raw experimental data and present them clearly.
3. Fit mathematical curves to material behaviors, and extract key properties using numerical integration and differentiation.

Course Content

A. Laboratory exercises for Materials Science and Engineering

1. Crystallography and crystal structures: Visualize and calculate atomic packing by using ball-and-stick models to construct Bravais lattices and identify Miller indices.
2. X-ray diffraction and Bragg's law: With raw XRD data files, extracting peak positions, applying Bragg's law, and indexing the peaks to identify the crystal system and lattice parameters.
3. Metallography and microstructural analysis: Observing and calculating the average grain size of different metal specimens (e.g., low carbon steel vs. brass) using optical microscope.
4. Hardness testing and defect strengthening: Performing Vickers hardness tests on various materials (metals, ceramics, polymers) and comparing the hardness.
5. Stress-strain analysis: Using a Universal Testing Machine (UTM), metal and polymer samples are fractured. Students extract the raw force-displacement data and plot the engineering stress-strain curves, determining Young's modulus, yield strength, ultimate tensile strength (UTS), and ductility.
6. The processing-property correlation -1: Ductile metal is cold rolled to obtain different thicknesses, measuring the increase in hardness.
7. The processing-property correlation -2: Cold-rolled sample is annealed and hardness is measured again.
8. Electrical conductivity: Differentiate between conductors and semiconductors. Using a four-probe setup, the electrical resistivity of a metal and a semiconductor is measured.
9. Band structure: Calculating the activation energy or band gap for the semiconductor by plotting log(conductivity) vs. 1/T.
10. Magnetic hysteresis: Characterize hard vs. soft magnetic materials. Generating hysteresis curves for different magnetic cores (e.g., iron, steel, ferrite).
11. Optical properties (absorption and bandgap): Using a UV-Vis spectrophotometer, students measure the absorption spectra of transparent polymers and a semiconductor powder suspension.

B. Laboratory Exercises for Numerical Methods for Materials Engineering:

1. Read a provided materials dataset (e.g., tensile test results or composition–hardness table) from CSV/Excel. Clean units, compute mean, variance, standard deviation, plot histograms and scatter plots; annotate with basic statistics.
2. Implement linear and polynomial regression for Arrhenius and Hall–Petch data; plot fits and residuals.
3. Nonlinear regression for Ramberg–Osgood stress–strain and diffusion
4. Compute work of deformation from tensile curves using trapezoidal vs Simpson's rule; compare errors and discuss choice of step size / sampling.
5. Integrating heat capacity to compute enthalpy or temperature-dependent properties.
6. Implement Euler and Runge–Kutta schemes for (a) lumped-capacitance cooling, and/or (b) simple first-order reaction kinetics; compare step sizes and methods.
7. Implement a 1D finite-difference scheme for diffusion or heat conduction, visualise evolution and discuss stability
8. Use finite-difference formulas (forward, backward, central) to compute $d\sigma/d\varepsilon$ from discrete stress–strain data, and dD/dT from diffusion data; study sensitivity to noise and the effect of smoothing.
9. Formulate small Ax=b systems from mixture rules (e.g., phase fractions), simple thermal networks or coupled force-balance problems; solve using built-in linear solvers and compare with hand calculations.

10. Demonstrate an ill-conditioned system and show how small changes in data affect the solution.

Course Material

For both labs, specific laboratory protocols and handouts will be made available for different experiments.

Reference Books

1. W. D. Callister Jr. and D. G. Rethwisch, Materials Science and Engineering: An Introduction, 10th ed. Hoboken, NJ, USA: John Wiley & Sons, 2020.
2. M. L. Boas, Mathematical Methods in the Physical Sciences, 3rd Edition, Wiley, 2006.

Title	**Kinetics of Materials**	Number	MTL 2XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite	Thermodynamics of Materials, Introduction to Materials Science and Engineering		

Objectives

1. To develop a fundamental understanding of kinetics in materials, focusing on diffusion, reaction rates, interface phenomena, and microstructural evolution.
2. To enable application of these principles to analyze and apply kinetic principles to materials processing, performance, and degradation across diverse material systems.

Learning Outcomes

1. Explain the fundamental concepts of kinetics, including driving forces, activation energy, and rate processes in materials.
2. Interpret and analyze transport, reaction, and interface-controlled phenomena using appropriate kinetic models and equations.
3. Apply kinetic principles to real materials processing scenarios such as sintering, additive manufacturing, and environmental interactions.

Contents

Fundamentals of Kinetics [6 lectures]: Thermodynamics vs kinetics, driving force and atomic mobility, energy barriers and activation energy, heterogeneous and homogeneous reactions, nucleation kinetics, overview of rate processes in materials, rate equations and temperature dependence, Arrhenius equation.

Reaction Kinetics in Materials Systems [7 lectures]: Rate laws: zero, first, second order reactions, effect of concentration and temperature on the reaction rate, homogeneous vs heterogeneous reactions, heterogeneous reaction kinetics: gas-solid, solid-liquid, liquid-liquid and solid-solid systems, catalysis and surface reaction kinetics.

Diffusion and Mass Transport [10 lectures]: Atomistic mechanisms: vacancy, interstitial, collective diffusion, random walk and diffusivity, Fick's first and second laws, analytical solutions (semi-infinite, thin films), diffusion in multicomponent systems (qualitative), role of defects: grain boundaries, dislocations, surfaces, experimental measurement of diffusion, case studies: carburization, doping in semiconductors, chemical potential and its role on diffusion flux, uphill and spinodal decomposition.

Interface Kinetics [5 lectures]: Interface structure and mobility, interface-controlled vs diffusion-controlled processes, moving boundary problems, Role of capillary for on kinetics, interfaces, its morphology, surface tension, oxidation kinetics (linear, parabolic laws), thin film growth basics (PVD/CVD).

Microstructural Evolution [6 lectures]: Driving forces for microstructural change (curvature, stored energy), phase transformations kinetics: concept of Johnson-Mehl-Avrami equation, precipitation, coarsening, recovery and recrystallization, grain growth kinetics.

Kinetics in Materials Processing [5 lectures]: Sintering kinetics and densification, diffusion and reactions in degradation processes, kinetics in additive manufacturing/rapid solidification, case studies linking diffusion, reaction, and microstructure.

Text Books

1. Materials Kinetics Fundamentals-Principles, Processes and Applications: Ryan O'Hayre, Wiley, 2015.
2. Physical Chemistry of Metals: L. Darken and R.W. Gurry, McGraw-Hill Book Company, Japan, 2002.
3. Chemical Kinetics: K. J. Laidler, McGraw-Hill (Pearson Education India), 2008.

Reference Books

1. Kinetics of Materials: R. W. Balluffi, S. M. Allen, W. C. Carter, Wiley, 2005.
2. Problems in Metallurgical Thermodynamics and Kinetics: G. S. Upadhyaya and R. K Dube, Pergamon Press, New York, 1982.
3. Phase Transformations in Metals and Alloys: D. A. Porter, K. E. Esterling, M. Y. Sherif, CRC Press, 2009.

Online Course Material

1. Thermodynamics and kinetics of materials: Dr. Saswata Bhattacharya, Material Science and Metallurgical Engineering, IIT Hyderabad. https://nptel.ac.in/courses/113106109

Title	**Materials for Functional Applications**	Number	MTL 3XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Objectives

1. To provide fundamental understanding of the classification, properties, and applications of functional materials in modern technology.
2. To develop knowledge of synthesis, characterization techniques, and the integration of functional materials into real-world devices.

Learning Outcomes

The students will be able to:

1. Analyse and correlate the structure–property–performance relationship of functional materials and evaluate suitable synthesis and characterization techniques.
2. Explain the working principles of electronic, optical, and magnetic materials and assess their applications in modern devices.

Contents

Introduction to Functional Materials (5 Lectures): Definition and classification of functional materials, Structure–property–performance relationship, Overview of advanced materials vs conventional materials, Role of functional materials in modern technology (electronics, energy, sensors, healthcare)

Synthesis and Growth Techniques (8 Lectures): Thin film growth and nanomaterial synthesis fundamentals, Top-down and bottom-up approaches, Solid-state synthesis, Chemical methods, Physical methods, stimuli responsive systems, Techniques for controlled molecular weight and dispersity.

Characterization Techniques (8 Lectures): Structural characterization, Morphological analysis, Spectroscopic techniques, Surface and compositional analysis, Electrical and thermal characterization

Electronic and Magnetic Functional Materials (5 Lectures): Basics of Band theory, semiconducting materials, magnetic materials, Thermoelectric materials, Piezoelectric and multiferroics

Optical and Photonic Materials (5 Lectures): Optical properties of materials, Photoluminescence and electroluminescence, Lasers and optical fibres, Materials for LEDs, Photodetectors and Solar Cells

Functional Materials in Devices (8 Lectures): 2D materials and their hybrids, Nanocomposites and multifunctional materials in devices, Case studies of real-world devices integrating functional materials

Text Books

1. Callister, W.D. and Rethwisch, D.G., (2020), *Materials Science and Engineering: An Introduction*, 10th Edition, Wiley.
2. Kasap, S.O., (2017), *Principles of Electronic Materials and Devices*, 4th Edition, McGraw-Hill Education.
3. Cao, G. and Wang, Y., (2011), *Nanostructures and Nanomaterials: Synthesis, Properties and Applications*, 2nd Edition, World Scientific.

References

1. Shackelford, J.F., (2014), Introduction to Materials Science for Engineers, 8th Edition, Pearson.
2. Poole Jr., C.P. and Owens, F.J., (2003), Introduction to Nanotechnology, Wiley-Interscience.
3. West, A.R., (2014), Solid State Chemistry and Its Applications, 2nd Edition, Wiley.
4. Sze, S.M. and Ng, K.K., (2006), Physics of Semiconductor Devices, 3rd Edition, Wiley-Interscience.

Online Course Material

1. NPTEL: Introduction to Materials Science and Engineering, IIT Kanpur, https://nptel.ac.in
2. NPTEL: Nanomaterials and Nanotechnology, IIT Delhi, https://nptel.ac.in
3. NPTEL: Electronic Materials and Devices, IIT Madras, https://nptel.ac.in

Title	**Materials for Energy Conversion and Storage**	Number	MTL 2020
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Objectives

The course outlines in a nutshell:

1. Different modes of energy conversion and storage
2. Development of materials with time and different applications, scale, energy and power density

Learning Outcomes

The students are expected to gain knowledge on:

1. Structure processing property performance relationship of the energy materials developed over time.
2. Different modes of energy conversion and storage with respect to scale, cost, energy and power density

Contents

Introduction [2 lectures]: Classification, Overview of Electrical (static), Electrochemical, Thermal, Mechanical, Photo-electrochemical (includes biological) energy storage.
Electrical (static) [4 lectures]: Capacitor, Leyden Jar, corona discharge, dielectrics.
Electrochemical conversion and storage [11 lectures]: Introduction to Electrolytic capacitor, Electrochemical cell (Battery), Supercapacitor, primary and secondary batteries, Li-ion battery, fuel cells.
Photo-electrochemical conversion [4 lectures]: Photo-electric effect, artificial photo synthesis, semiconductor based solar cells, dye-synthesized solar cells.
Thermal conversion and storage [8 lectures]: Solar thermal energy conversion, sterling engines, Molten salt technology, eutectic mixtures, Phase-change materials, thermoelectric materials.
Mechanical storage [6 lectures]: Pumped storage, gravity batteries, Compressing air storage, flywheels.
Carbon Materials for Energy conversion and storage [4 lectures]: Hybridization and carbon allotropes overview, Synthesis and characterization of carbon allotropes, Functionalization of carbon materials, Carbon nanomaterials for energy conversion and storage.

Textbook

1. Cheong K.Y., Impellizzeri,G and Fraga,M.A., (2018), *Emerging Materials for Energy Conversion and Storage* 1st Edition, Elsevier

Reference Book

1. Rosen, M.A., (2012) *Energy Storage*, Nova

Online Course Materials

1. Prof. Bazant M., MITopencourseware: Electrochemical Energy Systems, MIT Course Number 10.626 / 10.426, https://ocw.mit.edu/courses/chemical-engineering/10-626-electrochemical-energy-systems- spring-2014/
2. Prof. Pal K., Selection Of Nanomaterials For Energy Harvesting And Storage Application, NPTEL Course Material, Mechanical Engineering, Indian Institute of Technology Roorkee, https://nptel.ac.in/courses/112/107/112107283/

Title	**Transport Phenomena in Materials**	Number	MTL 2XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite	None		

Objectives

1. Develop understanding of the concepts of momentum, heat and mass transfer with a focus on transport processes in materials engineering.
2. Understand applications of transport phenomena in diverse material processes.

Learning Outcomes

1. Apply mass, momentum, and energy conservation laws to transport processes in materials.
2. Analyze and evaluate multiphysics systems across diverse material processes.

Contents

Basic concepts in Transport Phenomena [5 lectures]: Introduction of Transport Phenomena, Examples of Transport Phenomena in Materials Processing, Mathematical foundations, Control volume formulation, mass, energy and momentum balance and governing equations, development of continuity (conservation) equations. Dimensional analysis in transport processes - introducing key dimensionless parameters.
Momentum Transport [8 lectures]: Newton's Law of viscosity, Navier-Stokes equations and exact solutions for simple geometries, Friction factors and empirical relations in fluid flow, Unsteady state viscous flow, Transport in turbulent flow.
Heat Transport [8 lectures]: Governing equations for heat transfer and energy fluxes, Exact solutions for heat transfer problems, Equations for free and forced convection, Unsteady state heat conduction, Empirical correlations and heat transfer coupled with fluid flow.
Mass Transport [8 lectures]: Mass balance and governing equations for multicomponent systems, Diffusive mass transfer and exact solutions for simple geometries, Unsteady state mass diffusion, Solute transfer during phase change.
Coupled transport and applications [5 lectures]: Coupled momentum, energy and mass transport, Convective mass transfer correlations, Momentum, energy, and mass transport in boundary layers, Interphase momentum, heat, and mass transfer.
Applications and case studies [5 lectures]: Applications in metal, alloy and polymer processing, additive manufacturing, crystal and thin film growth, electron and charge transport in electronic and electrochemical systems. Simulation techniques in transport phenomena.

Textbooks

1. Bird, R. B., Stewart, W. E., Lightfoot, E. N., & Klingenberg, D. (2015). *Introductory Transport Phenomena, 1st edition,* John Wiley & Sons.
2. Gaskell, D. R., & Krane, M. J. M. (2024). *An introduction to transport phenomena in materials engineering, 3rd edition*, CRC Press.

References

1. Bird, R.B., Stewart, W.E. and Lightfoot, E.N. (2006). *Transport Phenomena, 2nd Edition,* John Wiley & Sons.
2. Poirier, D. R., & Geiger, G. H. (2016). *Transport phenomena in materials processing*. 1st *Edition,* Minerals, Metals & Materials Society.

Online Course Material

1. Phanikumar, G. (2025). *Transport Phenomena in Materials* [Online course]. NPTEL, IIT Madras. https://onlinecourses.nptel.ac.in/noc25_mm30/preview

2. Powell, Adam. *3.185 Transport Phenomena in Materials Engineering*. Massachusetts Institute of Technology: MIT OpenCourseWare, https://ocw.mit.edu.

Title	**Mechanical Behaviour of Materials**	Number	MTL 3020
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Objectives

1. To provide background on the mechanisms of deformation in engineering materials, and on the evolution of material microstructure when subjected to mechanical stress during materials processing and engineering applications

Learning Outcomes

1. Ability to design materials with improved mechanical properties and identify failure mechanisms in materials for engineering applications
2. To measure important mechanical properties and correlate with the microstructure

Contents

Continuum Theory of Deformation [2 lectures]: Stress and strain tensor, Generalized Hooke's Law, Yield *Criteria Lattice Defects [6 lectures]:* Vacancies and interstitials, dislocations: Edge and Screw, stress fields around edge and screw dislocations, strain energy of dislocations, force on dislocation, dislocation-dislocation interaction

Plastic Deformation [7 lectures]: Slip in a perfect lattice, slip by dislocation movement, Critical Resolved Shear stress, cross-slip and climb, deformation of single crystals, dislocations in different crystal structures, twinning

Strengthening Mechanisms [6 lectures]: Grain boundary strengthening, strain hardening, solid-solution hardening, age-hardening, martensite strengthening

Fracture and Toughening Mechanisms [6 lectures]: theoretical fracture strength and Griffith's Law, modes of fracture, fracture toughness, brittle and ductile fracture, toughening mechanisms in metals, ceramics, and polymers

Fatigue and Creep [6 lectures]: Introduction to fatigue, stress and strain-life approach, cyclic stress-strain curve, fatigue crack propagation, high temperature deformation, fatigue-creep interaction

Deformation of Polymers [6 lectures]: Elastic deformation in semi crystalline polymers, nonlinear elastic deformation, shear banding and crazing, stress-strain curves

Text Books

1. Courtney, T. H., (2005), *Mechanical Behaviour of Materials*, 2nd Edition, Waveland Press Inc.
2. Dieter, G.E., (1989), *Mechanical Metallurgy*, SI Metric Edition, McGraw-Hill Book Company.
3. Shetty, M. N., (2013), *Dislocations and Mechanical Behavior of Materials*, Prentice Hall India Learning Private Limited

Reference Books

1. Hertzberg, R.W., (2013), *Deformation and Fracture Mechanics of Engineering Materials,* 5th Edition, Wiley
2. McClintock,F.A., and Argon,A.S., (1966), *Mechanical Behavior of Materials*, Addison-Wesley.

Online Course Material

1. Van Vilet, K., *3.22* Mechanical Behavior of Materials, Spring 2008. Massachusetts Institute of Technology: MIT OpenCourseWare, https://ocw.mit.edu/courses/materials-science-and-engineering/3- 22-mechanical-behavior-of-materials-spring-2008/

Title	**Materials for Functional Applications Lab**	Number	MTL 2XX0
Department	Materials Engineering	L – T – P [C]	0 – 0 – 4 [2]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Course Content

A. Laboratory exercises for Materials for Functional Applications

1. Synthesis of Graphene/Molybdenum disulphide via Sono chemical Exfoliation
2. Optical Microscopy: Identification of layer thickness and surface uniformity of 2D materials using optical contrast.
3. Raman Spectroscopy: Analysis of vibrational modes to confirm material quality, defects, and number of layers.
4. UV–Visible Spectroscopy: Study of optical absorption and bandgap estimation of materials.
5. Electrospinning: Synthesis of Polyacrylonitrile (PAN) nanofibers.
6. Morphological characterization of Polyacrylonitrile (PAN) nanofibers.
7. Synthesis of Nylon66 by interfacial polymerization
8. Fabrication of low-density polyethylene/Polyvinyl alcohol with functional fillers
9. Synthesis of absorber layer for the solar cells.
10. Structural, optical and electrical characterization of the absorber layer for the solar cells.

B. Laboratory Exercises for Energy Materials:

1. Demonstration of static electricity storage with electrostatic generator, Corona effect, Energy storage in Leyden jar. Fabrication of a capacitor
2. Charging and discharging of a capacitor, Measuring the capacitance of a Capacitor, Role of capacitor in voltage stabilizing/regulating, filtering
3. Fabrication and operation of a Lead-acid battery, Charge discharge voltammetry/EIS characteristics of Lead-acid battery
4. Fabrication and operation of a Lithium-ion battery, half-cell and full cell configuration, Charge discharge/voltammetry/EIS characteristics of Lithium-ion battery
5. Degradation test for lithium ion batteries, Use of different cathodes, Cell fabrication and dismantling, observation of structural changes, effect of temperature
6. Battery aging and self-discharge of lead-acid and lithium ion battery, battery for electric vehicles
7. Conversion of carbon dioxide into propane through artificial photosynthesis
8. Experimental Evaluation of a Paraffin as Phase Change Material for Thermal Energy Storage
9. Modelling and validation of a flywheel energy storage, modelling of gravity assisted storage systems (can be taken before)
10. Synthesis of different forms of carbon nanomaterials for capacitor, artificial photosynthesis devices and electrodes for batteries, observation of different morphologies and structural reversibility (demonstration experiment, make it simple)

Course Material

For both labs, specific laboratory protocols and handouts will be made available for different experiments.

Reference Books

1. Cheong K.Y., Impellizzeri, G and Fraga, M.A., (2018), Emerging Materials for Energy Conversion and Storage 1st Edition, Elsevier
2. Cao, G. and Wang, Y., (2011), Nanostructures and Nanomaterials: Synthesis, Properties and Applications, 2nd Edition, World Scientific.

Title	**Materials Degradation**	Number	MTL 3XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Objectives

1. Identify the various forms of corrosion and their mechanisms.
2. Establish the thermodynamic driving forces and kinetic barriers that govern metallic degradation.
3. Develop proficiency in using mixed potential theory to predict corrosion rates.

Learning Outcomes

1. Develop the understanding of basic electrochemical processes and redox reactions.
2. Design protection strategies using electrochemical methods, alloy selection, and advanced surface coatings.

Contents

Introduction [3 lectures]: Corrosion – definition, different forms of environmental degradation, cost of corrosion, electrochemical nature.

Thermodynamics of Corrosion [8 lectures]: Free energy and electrochemical potential – EMF series, Nernst relationship, important reactions, cell potential, reference electrodes, Pourbaix diagram and its importance in corrosion, construction of Pourbaix diagrams for Al, Cu, Fe, and Ni.

Kinetics of corrosion [6 lectures]: Current density and corrosion rate, corrosion rate expressions, exchange current density, polarization – activation, concentration and resistance overpotential.

Mixed potential theory [9 lectures]: Mixed potential theory, applications to active metals, passivation, application of mixed potential theory in passivating metals, understanding of corrosion on the basis of mixed potential theory: activation controlled and concentration controlled corrosion.

Corrosion protection [4 lectures]: Cathodic protection, sacrificial anode and impressed current cathodic protection (ICCP), anodic protection, industrial application and calculations, protection of metals and alloys, coatings, inhibitors, materials selection/substitution.

Corrosion testing [3 lectures]: Immersion test, salt fog test, electrochemical work bench-3-electrode electrochemical cells, reference electrode, counter electrode and potentiostat. Potentiodynamic test, Galvanostatic test, linear polarization.

Stress effect on corrosion [3 lectures]: stress corrosion cracking (SCC), corrosion fatigue, hydrogen embrittlement.

Thermodynamics of oxidation [3 lectures]: oxidation of metals and alloys, Pilling Bedworth ratio, Ellingham diagram, kinetics of oxidation, effect of doping on the oxidation behavior.

Textbooks

1. Jones, Denny A., *Principles and Prevention of Corrosion*, 2nd Ed., Pearson.
2. Fontana, Mars G., *Corrosion Engineering*, 3rd Ed., McGraw-Hill. (The industry standard for practical failure analysis).

References

1. Branko N. Popov, (2006), *Corrosion Engineering: Principles and Solved Problems*, 1st Edition, Elsevier
2. Revie, R. Winston, and Uhlig, Herbert H., *Corrosion and Corrosion Control*, 4th Ed., Wiley.

Online course material

1. Prof. Kallol Mondal, *Corrosion – Part 1* (NPTEL) https://nptel.ac.in/courses/113104082
2. Prof. Kallol Mondal, *Corrosion – Part 2* (NPTEL) https://onlinecourses.nptel.ac.in/noc19_mm15/preview
3. V.S. Raja, Aqueous corrosion and its control (NPTEL) https://nptel.ac.in/courses/113101098

Title	**Phase Transformations**	Number	MTL 3030
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Objectives

The Instructor will:

1. Explain the importance of phase transformations in solid and liquid state and their applications through experimentation

Learning Outcomes

The students are expected to have the ability to:

1. Understand the microstructure evolution during the phase transformation
2. Understand the variation of properties with the microstructure evolution towards developing heat treatment-microstructure-property relationship

Contents

Thermodynamics and Kinetics [6 Lectures]: free energy, order of transformation, driving force, homogeneous and heterogeneous nucleation, growth kinetics, coarsening, precipitation
Phase Transformation during Solidification [4 Lectures]: Eutectic and Peritectic Transformation
Heat-Treatment of Ferrous and Nonferrous Metals [4 Lectures]: Annealing, Normalizing, Hardening and Tempering, Precipitation Hardening
Interfaces [4 Lectures]: atomic mechanisms of diffusion, activation energy, interfacial free energy, types of interface (coherent, semi-coherent and incoherent interfaces), interface migration
Diffusional Phase Transformations in Solids [14 Lectures]: austenite, transformation of austenite, TTT diagram, eutectoid transformation, pearlite and bainite transformation, order-disorder transformation, precipitation hardening, spinodal decomposition and massive transformation
Diffusionless Phase Transformations in Solids [7 Lectures]: *Martensite transformation:* characteristics and nature, morphology, crystallography, theory of nucleation and growth, and pre-martensite phenomena, martensitic transformation in steel. Martensite in non-Ferrous systems-thermoelastic martensite and shape memory effect.

Textbooks

1. Raghavan,V., (1987), *Solid State Phase Transformations*, 1st edition, Prentice Hall India,.
2. Porter,D.A., Easterling,K.E., and Sherif,M.Y., (2009), *Phase Transformation in Metals and Alloys*, 3rd edition, CRC Press,
3. Abbaschian,R., Abbaschian,L., and Reed-Hill,R.E., (2009), *Physical Metallurgy Principles*, 4th edition, Cengate Learning,
4. Rajan,T.V., Sharma,C.P., and Sharma,A., (2011), *Heat Treatment: Principles and Techniques*, 2nd edition, Prentice Hall India

Reference Books

1. Khachaturyan,A.G., *Theory of Structural Transformations in Solids*, 1st edition, Dover publications, 2008.
2. Avner, S.H., *Introduction to Physical Metallurgy*, 2nd edition, McGraw Hill Education, 2017

Online Course Material

1. Gururajan,M.P., *Phase Transformations and Heat Treatment*, NPTEL Course Material, Department of Metallurgical Engineering and Materials Science, Indian Institute of Technology Bombay, http://nptel.ac.in/courses/113101003/

Title	**Materials Processing and Manufacturing**	Number	MTL 3060
Department	Materials Engineering	L–T–P [C]	3–0–0 [3]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Objectives

The instructor will:

3. Introduce the various industrially relevant materials processing and manufacturing techniques.
4. Impart fundamental understanding of various casting, welding and forming processes.

Learning Outcomes

The students are expected to have the ability to:

3. Understand how material properties vary with microstructural changes induced by processing and manufacturing towards developing microstructure-property-processing relationship.

Contents

Solidification [6 lectures]: Thermodynamics of solidification, pure metal solidification, Alloy solidification: Constitutional undercooling, nucleation, dendritic growth.

Bulk Processing of Materials [14 lectures]: Casting: Structure of casting and ingots, types of casting process, casting defects. Welding: Classification of welding processes, welding defects, Brazing and Soldering, Microstructural mechanisms associated with metals joining operations. Powder metallurgy. Additive manufacturing of metals.

Bulk Metal Forming Techniques [13 lectures]: Rolling: Cold and hot rolling, mechanics of rolling, microstructural changes during rolling, rolling mill, rolling defects. Forging: Open and closed die forging, upsetting and heading, forging defects. Extrusion: Types of extrusion, metal flow in extrusion. Wire and rod drawing: Die design and materials, optimal die angle. Factors affecting metal working.

Sheet Metal Forming Techniques [6 lectures]: Sheet metal forming: Bending, stretching and deep drawing, compound and progressive die, plastic anisotropy parameter, forming limit diagram, microstructural changes.

Textbooks

1. Flemings, M.C., (1974), *Solidification Processing*, Mcgraw Hill Book Company.
2. Dieter, G.E., (1986), *Mechanical Metallurgy*, 3rd Edition, McGraw Hill Book Company.
3. Ghosh, A. and Mallik, A. K., (2017), *Manufacturing Science*, East-West Press.

Reference Books

1. Chalmers, B., (1967), *Principles of Solidification*, Wiley.
2. Stefanescu, D.M., (2015), Science *and Engineering of Casting Solidification*, 3rd Edition, Springer.
3. Hosford, W.F., Caddell, R.M., (2014), *Metal Forming: Mechanics and Metallurgy*, 4th Edition, Cambridge University Press.

Online Course Material

1. Shekhar, S., Fundamentals of Materials Processing (Part–1), NPTEL Course Material, DMSE, IIT Kanpur, https://nptel.ac.in/courses/113/104/113104073/
2. Shekhar, S., Fundamentals of Materials Processing (Part–2), NPTEL Course Material, DMSE, IIT Kanpur, https://nptel.ac.in/courses/113/104/113104075/

Title	**Synthesis and Processing of Semiconductor materials**	Number	MTL 3050
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Objectives

The instructor will:

1. Introduce different synthesis and fabrication techniques used for semiconductor device fabrication.
2. Introduce different synthesis techniques used for 2D materials

Learning Outcomes

After completion of the course students will be have a basic understanding of different semiconductor device fabrication techniques as well as gain knowledge how to synthesize various 2D materials

Contents

Semiconductor processing [16 Lectures]: Silicon Wafer Fabrication Physical vapour deposition methods: e-beam, pulsed laser and ion beam evaporation, Glow, Discharge and Plasma, Sputtering–mechanisms and yield, DC and RF sputtering, Chemical vapour deposition: Sol-gel, Spin Coating, Atomic Layer deposition, CVD, PECVD, MOCVD, solution processing techniques, (printing, spin coating, chemical bath depositions, etc.). Semiconductor Integration with Functional oxides and Challenges in fabrication

2D Materials Processing [Lectures 13]: Different types of 2D materials, Strategies and challenges of 2D materials synthesis: micromechanical exfoliation, ultrasonic exfoliation, hydrothermal method, topochemical transformation, chemical vapor deposition, Top Down vs Bottom up, Production of graphene and 2-d materials: Comparison of production methods, Scotch-tape method (micromechanical cleavage), Chemical vapour deposition, Solution-exfoliation 1 – graphene and other 2-d materials, Solution-exfoliation 2 – graphene oxide, Decomposition of silicon carbide, Production of graphene nano-ribbons, Synthesis of hexagonal boron nitride, and transition metal dichalcogenides, to new and emerging materials such as black phosphorus, silicene, and germanene.

Lithography and Etching [Lectures 6]: Pattern Formation, Optical Lithography, Resists, Wet Chemical Etching, Reactive Ion Etching, Ion implantation doping

Metallization and Polishing [Lectures 2]: Ohmic and Schottky contacts, contact formation and interconnects in semiconductor devices.

Cleanroom & Contamination [Lectures 2]: Cleanroom standards, Contamination and yield, Environment, safety and health

Textbook

1. Ohring, M., (2002), The materials science of Thin films, 2nd Edition, Academic Press Ltd.
2. Chopra, K. L., (1969), Thin Film Phenomena, McGraw-Hill Book Company

Reference Books

1. Editor Gogotsi, Y., (2019), 2D Metal Carbides and Nitrides (MXenes): Structure, Properties and Applications, Springer
2. Shinohara, H., Tiwari, A., (2015), Graphene: an introduction to the fundamentals and industrial applications. John Wiley & Sons

Online Course Material

1. Parasuraman Swaminathan, Electronic materials, devices and fabrication, NPTEL Course Material, Department of Metallurgical and Materials Engineering, IIT Madras, https://nptel.ac.in/courses/113/106/113106062/

Title	**Computational Modelling of Materials**	Number	MTL 3050
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Objectives

1. Introduce Phase-Field Modelling in the context of Materials Engineering
2. Introduce the theory behind the first-principles calculations
3. Create awareness of the opportunities and limitations of Molecular dynamics simulations

Learning Outcomes

1. Ability to formulate model and solve initial value and boundary value problems
2. Understand the working principles of the first-principles calculations and molecular dynamics
3. Simulate materials using the first-principles calculations and molecular dynamics

Contents

Phase-Field Modelling [13 lectures]: Microstructure evolution, Continuum vs sharp interface description, The Ginzburg-Landau free energy functional, Equilibrium interfaces and surface tension, Conserved and non-conserved order parameters, Driving forces, fluxes, Spinodal decomposition in a binary alloy

Molecular Dynamics [13 lectures]: Molecular dynamics: Applications, basic concepts, potential energy functions, integration algorithms, temperature control, boundary conditions, neighbour lists, initialization, equilibrium, geometry optimization, static properties, dynamical properties, analysis of results, visualisation

First Principle Calculations [13 lectures]: Notation & Mathematics: Dirac notation, Fourier transformation, Function and functional, Quantum mechanics: Schrodinger equation, Born-Oppenheimer approximation, Mean-field methods, Kohn-Sham method: Density functional theory, Hohenberg-Kohn theorems, Kohn-Sham method, Exchange correlation functional, Structure relaxation, Smearing methods, Electronic band structure, Fermi-Dirac distribution

Text Books

1. Provatas, N., Elder, K., Phase-Field Methods in Materials Science and Engineering, John Wiley & Sons, 2011.

2. Frenkel, D., Smit, B., *Understanding Molecular Simulation: From Algorithms to Applications*, 2nd Edition, Elsevier 2001.

3. Lee, J. G., Computational Materials Science: An Introduction, 2nd edition, CRC Press, 2016.

References

1. Sholl, D. S., and Steckel, J. A., *Density Functional Theory: A Practical Introduction*, 1st Edition, Wiley, 2009.
2. Parr, R.G., and Yang, W., *Density-Functional Theory of Atoms and Molecules*, 1st Edition, Oxford Science Publications, 1994.
3. Lesar, R., *Introduction to Compuational Materials Science*

Online Course Material

1. Ceder, G. and Marzari, N., 3.320 *Atomistic Computer Modeling of Materials* (SMA 5107). Spring 2005. Massachusetts Institute of Technology: MIT OpenCourseWare, https://ocw.mit.edu. License: Creative Commons BY-NC-SA

Title	**Physical Metallurgy and Materials Modelling Lab**	Number	MTL 2XX0
Department	Materials Engineering	L – T – P [C]	0 – 0 – 4 [2]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Objectives

To introduce the students to

1. Different types of phase transformations in solid and liquid state experiments, and to provide hands-on training on various mechanical testing equipment to measure mechanical properties of materials.
2. Hands-on training in running simulations related to materials science and engineering.

Learning Outcomes

The students are expected to have the ability to

1. Apply the knowledge of phase transformation to understand microstructure and mechanical property evolution in materials.
2. Measure mechanical properties of materials.
3. Simulate materials using numerical methods, molecular dynamics, first-principles calculations

Course Content

A. **Laboratory Experiments of Physical Metallurgy Lab**

1. Uniaxial monotonic stress-strain response of metals, ceramics, and polymers
2. Hardness via spherical and Vickers indentation and correlation with yield strength
3. Effects of annealing, normalizing, and hardening heat treatments on microstructure and mechanical properties of steels
4. The effects of grain size on yield strength
5. Effects of cold working and subsequent annealing on microstructure and mechanical properties 6. Hardenability of steels using Jominy end quench test
7. Age hardening study of a range of aerospace materials (Al-alloys, Ti-alloys, superalloys)
8. Activation energy and Stored energy calculation for phase transformation
9. Measurement of quasi-static and indentation fracture toughness
10. Measurement of impact fracture toughness using Charpy/Izod impact test
11. Fracture surface interpretation using SEM
12. Creep of a 60%Sn-40%Pb Solder wire at room temperature

B. **Laboratory Experiments if Computational Modeling of Materials Lab**

Phase field modeling exercises

1. Spinodal decomposition simulation by solving Cahn-Hilliard Equation
2. Modeling grain growth with Allen-Cahn equation.
3. Modeling of solidification process (dendritic growth).

Molecular dynamics Exercises

1. Calculate the energy in different crystal lattice structures
2. Determination of coefficient of thermal expansion
3. Application of tensile load and visualize the results
4. Determination of bulk modulus

First-principles calculations Exercises

1. Preparation of input files and visualisation of structures
2. Optimisation of simulation parameters.
3. Determination of equilibrium lattice parameter
4. Structure optimisation using a variable-cell method

Course Material

For both labs, specific laboratory protocols and handouts will be made available for different experiments.

Reference Books

1. Lee, J. G., Computational Materials Science: An Introduction, 2nd edition, CRC Press, 2016.
2. *Provatas, N., Elder, K., Phase-Field Methods in Materials Science and Engineering, John Wiley & Sons, 2011.*
3. Porter,D.A., Easterling, K.E., and Sherif, M.Y., (2009), Phase Transformation in Metals and Alloys, 3rd edition,

Title	**Metallurgy of Joining**	Course Number	MTL6xxx
Department	Materials Engineering	L-T-P: C	3-0-0: 3
Offered for	B.Tech.	Type	Program Compulsory (PC)
Prerequisites			

Objectives:

1. To introduce the basics of materials joining techniques and the associated metallurgy of welds.
2. To develop an understanding of the relationship between joining processes, microstructure evolution, and mechanical properties in welded materials.

Learning outcomes:

The students are expected to have the ability to:

1. Understand the importance of the materials joining in structural applications.
2. Study the fundamentals of welding microstructures and properties

Contents

Materials Joining basics [6 lectures]: Materials joining basics, type of joints, Types of welding techniques, heat input and joining efficiency, importance of welding in structural applications

Fusion welding [8 lectures]: Important welding techniques such as submerged arc welding, flux core arc welding, resistance spot welding, tungsten inert gas welding, Metal inert gas welding, plasma arc welding, electron beam welding, laser beam welding.

Solid state welding [8 lectures]: Important welding techniques like Diffusion bonding, friction welding, friction stir welding, adhesive joining, ultrasonic welding, forge welding, industrial applications of solid-state welding.

Metallurgy of fusion welding [6 lectures]: Fusion zone, partially melted zone, heat affected zone, residual stress in welding, epitaxial growth during welding, heat transfer in welding, relationship between microstructure and properties, effect of cooling rates.

Metallurgy of solid-state welding [6 lectures]: materials plasticization, bonding interfaces, effect of processing parameters, heat input in welding, defects and mechanical properties of solid state welds.

Case Studies [5 lectures]: Welding in EV batteries, fusion welding of Al alloys, Friction stir welding of Mg alloys, Laser beam welding of Al alloys and steel

Textbooks

1. Kou S., (2003), Welding Metallurgy, 2rd Edition, Jhon Willey and Sons, New Jersey, United States
2. Lippold J.C., (2015) Welding Metallurgy and Weldability, Jhon Willey and Sons, New Jersey, United States

Reference

1. Reeser M.A., (2017), Welding complete: Techniques, project plans and instructions, cool springs press, MA, USA.
2. George E Linnert, Welding Metallurgy, 4th edition, GML Publications, USA.

Online

1. NPTEL lectures on weld metallurgy: https://nptel.ac.in/courses/113107092.
2. NPTEL lectures on welding processes: https://onlinecourses.nptel.ac.in/noc25_me149.

Title	**Materials Characterization**	Number	MTL 3XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Objectives

1. To understand the principle of the instruments and methods for characterizing different materials and their properties

Learning Outcomes

The students are expected to have the ability to:
1. Know the basic working principles of various instruments and use the same while performing experiments
2. Choose the characterization methods based on the requirement and performance

Contents

Introduction to material characterization [2 Lecture]: Phase, purity, and defects analysis of materials and its overview
Light Optical Microscopy [4 Lectures]: polarization microscopy, phase contrast microscopy, DIC Microscopy and confocal microscopy
Diffraction and Electron Microscopy Techniques [10 Lectures]: X-Ray Diffraction, Scanning electron microscopy (SEM) & Transmission electron microscopy (TEM)
Morphological Characterization [8 Lectures]: Atomic force microscopy (AFM) & Scanning Tunnelling Microscope
Spectroscopic Characterization [10 Lectures]: UV-VIS, FTIR, Raman spectroscopy and related case studies
Thermal Analysis [5 Lectures]: Differential scanning calorimetry (DSC), Thermal gravimetric analysis (TGA)

Textbook

1. Zhang,S., Li,L. and Kumar,A., (2008), Materials Characterization Techniques, , CRC Press
2. Cullity, B.D., Stock, S.R., (2014), Elements of X-Ray Diffraction, 3rd Edition, Pearson Education Limited

Reference Books

1. Evans,C., Brundle.R. and Wilson,(1992), Encyclopaedia of Materials Characterization: Surfaces, Interfaces, Thin Films (Materials Characterization Series), Butterworth-Heinemann,
2. Kaufmann,E.N., (2012), Characterization of Materials, 3 Volume Set, 2nd Edition, Wiley
3. Egerton,R., (2016), Physical Principles of Electron Microscopy: An Introduction to TEM, SEM and AEM, 2nd ed. Springer
4. Hwang,J-Y, Monteiro,S.N., Bai,C., Carpenter,J.S., Cai,M., Firrao,D., Kim,B-G., (2012), Characterization of Minerals, Metals and Materials, Wiley

Online Course Materials

1. Shankaran,S., Materials Characterization, NPTEL Course Material, Department of Metallurgical & Materials Engineering, Indian Institute of Technology Madras, http://nptel.ac.in/courses/113106034/
2. Alagarsamy,P., Characterization of Materials, NPTEL Course Material, Department of Physics, Indian Institute of Technology, Guwahati, http://nptel.ac.in/courses/115103030/
3. Biswas,K. and Gurao,N.P., NPTEL Course Material, Advanced Characterization Techniques, Department of Materials Science and Engineering, Indian Institute of Technology, Kanpur, http://nptel.ac.in/courses/113104004/

Title	**Principles of Metal Extraction**	Number	MTL 3XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Objectives

1. Outline the thermodynamic and kinetic principles that govern the metal extraction from ores.
2. Understand the thermal profile and chemical reactions within iron blast furnace, alternative iron making routes, and steel making processes.
3. Understand the reason behind the chemical processing routes (pyro, hydro, and electro) for non-ferrous metals.

Learning Outcomes

1. Understand the basics of process metallurgy involved in iron and steel making processes.
2. Be cognizant of advanced non-ferrous metal extraction processes in industries.

Contents

Introduction [10 lectures]: Important ores, sources and their applications, mineral beneficiation: comminution – crushing and grinding, classification – sieve analysis, concentration – gravity separation, magnetic separation, froth flotation, thermodynamic foundations of metal extraction, principles of pyrometallurgy, standard free energy calculations, Ellingham diagrams, predominance area diagrams, Pourbaix diagrams, Understanding unit processes in extractive metallurgy – calcination, roasting, leaching, precipitation, ion exchange, solvent extraction, smelting etc

Fundamentals of iron making [15 lectures]: concept of integrated steel plant, different types of ironmaking processes, agglomeration techniques, thermodynamics, kinetics and transport phenomena in iron and steelmaking processes, temperature, burden distribution, and aerodynamics of blast furnace, physical and chemical; processes in upper and lower zones of blast furnace, direct reduction process-DRI, HBI, smelting reduction process, Corex, Romelt, HiSmelt, Finex etc. Modern and secondary steel making, basic oxygen furnace steelmaking, electric arc furnace (EAF), and induction furnace steelmaking, deoxidation, ladle and tundish Metallurgy, continuous casting, casting defects, green processes of iron and steelmaking.

Extraction of non-ferrous metals [14 lectures]: General routes of non-ferrous extraction, electrometallurgical and hydrometallurgical routes, extraction of metals from oxide ores (Al, Mg, Ni), extraction of metals from sulphide ores (Cu, Pb, Zn, Ni), halide route of metal extraction (Ti, nuclear reactor metals, Na, Mg, and Ca), rare earth metals, precious metal extraction routes.

Textbooks

1. H. S. Ray, R. Sridhar, and K. P. Abraham, *Extraction of nonferrous metals*, Affiliated East West Press Pvt Ltd., New Delhi (2007)
2. A. Ghosh, and A. Chatterji, (2008), *Ironmaking and Steelmaking: Theory and Practice*, Prentice-Hall (India)
3. R. H. Tupkary and V. R. Tupkary (2000), *An Introduction to Modern Steel Making,* 7th edition, Khanna Publishers (India).

Reference

1. Dennis, W.H., *Extractive Metallurgy*, Philosophical Library, New York (1965)
2. Chatterjee,A., (1994), *Beyond the Blast Furnace*, CRC Press
3. R. H. Tupkary and V. R. Tupkary, (2000), *An Introduction to Modern Steel Making*, Khanna publishers (India.)

Online course material

1. Ray, H.S., Non-ferrous Extractive Metallurgy, NPTEL Course Material, Department of Materials and Metallurgical Engineering, Indian Institute of Technology Kharagpur, https://nptel.ac.in/courses/113/105/113105021/
2. Muzumdar,D., and Koria,S.C., Steelmaking, NPTEL Course Material, Department of Material Science and Engineering, Indian Institute of Technology Kanpur, http://nptel.ac.in/courses/113104013/
3. Gupta,G.S., Iron Making, NPTEL Course Material, Department of Materials Engineering, Indian Institute of Science Bangalore, http://nptel.ac.in/courses/113108079/

Title	**Machine Learning for Materials Informatics**	Number	MTL6XX0
Department	Materials Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech.	Type	Program Core (PC)
Prerequisite			

Objectives

1. To build complete ML pipelines for materials problems: data preparation, featurization, model selection, evaluation and interpretation.
2. To give hands-on experience with regression, classification and clustering for real materials datasets, using the algorithms learned earlier in the institute ML course.

Learning Outcomes

Students will be able to

1. Construct and clean materials datasets and build appropriate feature sets (composition, structure, microstructure, processing) and use unsupervised learning to explore materials spaces.
2. Train, validate and interpret ML models (regression and classification) for materials-property prediction and critically assess model performance, uncertainty for materials decisions.

Contents

Materials datasets and featurization (9 Lectures): Types of materials datasets; scalar, vector and image data. Data cleaning: missing values, outliers, unit consistency; basic exploratory analysis. Compositional descriptors: simple composition-based features (averages, differences, moments of atomic properties); introduction to established descriptor ideas at high level. Structural, microstructural and processing descriptors: space group, lattice parameters, phase labels, grain size, phase fractions, texture indicators; encoding process routes and heat-treatment schedules.

Supervised learning for materials (7 Lectures): Problem formulation: targets, features, data splits (train/validation/test), cross-validation; avoiding data leakage. Regression tasks: using linear regression, k-NN, tree-based models or simple ensembles to predict continuous properties. Classification tasks: binary/multi-class problems; metrics. Model selection and regularisation at an intuitive level.

*Unsupervised learning and exploration (4 Lectures):*Clustering: k-means and/or another simple algorithm; grouping alloys/ceramics into families based on descriptors. Dimensionality reduction: PCA for visualising high-dimensional descriptor spaces; interpreting principal components in materials terms.

*Interpretation, reliability and small-data issues (6 Lectures):*Feature importance and simple sensitivity analyses; interpreting learned relationships in the context of known materials science. Overfitting, domain of applicability, uncertainty and robustness; pitfalls with small and biased materials datasets. Documentation and reproducibility; basic good practices for datasets and models.

Deep Learning and Large Language Models for Materials Informatics (8 Lectures): Introduction to neural networks: perceptron, shallow and deep neural networks; basic ideas of training, activation functions, and backpropagation. Overview of deep learning architectures relevant to scientific and materials data. Introduction to transformer models and large language models (LLMs)- embeddings, attention mechanism, prompting, fine-tuning, and limitations of LLMs. Applications of LLMs in engineering and materials science, including literature mining, materials-property prediction, workflow automation, scientific assistants, inverse materials design, and autonomous laboratories.

Machine Learning Interatomic Potentials (MLIPs) (5 Lectures): Introduction to MLIPs and their importance in modern computational materials science. Descriptor concepts and representations for atomic environments. Overview of neural-network interatomic potentials, Gaussian Approximation Potentials (GAP), Equivariant Deep Descriptor Potentials (EDDPs)

Text Books

1. Krishna Rajan, Materials Informatics: Molecules, Crystals and Beyond, 1st Edition, Elsevier, 2026.
2. Olexandr Isayev, Materials Informatics - Methods, Tools, and Applications, Wiley India Pvt Ltd, 2018

Reference Books

1. C. M. Bishop, Pattern Recognition And Machine Learning, SPRINGER NP EXCLUSIVE(CBS), 2009.
2. Pascal T. SalzbrennerCorresponding Author ORCID logo ; Se Hun Joo ORCID logo ; Lewis J. Conway ORCID logo ; Peter I. C. Cooke ORCID logo ; Bonan Zhu ORCID logo ; Milosz P. Matraszek ORCID logo ; William C. Witt ORCID logo ; Chris J. Pickard, Developments and further applications of ephemeral data derived potentials, J. Chem. Phys. 159, 144801 (2023)

Online Course Materials

1. NPTEL – Introductory Materials Informatics (IIT Kanpur/ IIT Ropar). https://nptel.ac.in/courses/113104761

Title	**Processing of Polymers and Ceramics**	Number	MTL 3XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Objectives

1. Develop a thorough understanding of the major industrial processing routes used for polymers and ceramics.
2. Provide the scientific basis for the relationship between processing parameters, microstructure, and final material properties.

Learning Outcomes

1. Identify and explain the working principles of key polymer and ceramics processing techniques and able apply process selection criteria to choose appropriate manufacturing routes for a given material and application.
2. Analyse the effect of processing parameters on the microstructure and properties of polymeric and ceramic products.

Contents

Introduction to Polymer Processing [5 lectures]: Overview of polymers: thermoplastics, thermosets, elastomers, molecular weight and distribution, Synthesis route, Rheology, viscoelasticity and melt flow, processing-structure-property relationships, Classification of Polymer processing operations.

Extrusion and Moulding based Process[10 lectures]: Single screw extrusion, twin screw extrusion, film processing, Sheet, Co-extrusion, Injection molding, blow molding, compression molding and rotational moulding

Mixing and forming [6 lectures]: Compounding, Two-roll mill, Thermoforming, Calendaring, Reactive processing and reaction injection Molding, Composite Processing, Fiber reinforced plastics. Recycling

Introduction to Ceramics Processing [4 lectures]: Classification of ceramics: oxides, non-oxides, glasses; raw material selection and characterization; powder synthesis: solid-state, sol-gel, chemical vapor synthesis; powder characterization: size, morphology, surface area, equivalent diameter of powders, powder mixing (dry and wet), Rheology.

Forming of Ceramics [6 lectures]: Dry and semi dry pressing, uniaxial and isostatic pressing; slip casting and tape casting, extrusion of ceramic pastes, injection molding of ceramics, gel casting, colloidal processing; binder systems and their removal; green body characterization.

Sintering and Densification [4 lectures]: Driving forces for sintering; solid-state sintering: initial, intermediate and final stages; liquid-phase sintering; pressure-assisted sintering: hot pressing and hot isostatic pressing (HIP); spark plasma sintering; microstructure evolution during sintering; grain growth and densification maps.

Advanced Ceramic Processing [4 lectures]: Fundamentals of glass processing, Glass melting, refining, and forming processes, chemical vapor deposition (CVD) for ceramic coatings (thermal and plasma spray), ceramic matrix composites (CMC): fabrication and applications; quality control in ceramics manufacturing.

Text Books

1. Osswald, T.A. and Menges, G., (2012), Materials Science of Polymers for Engineers, 3rd Edition, Hanser Publishers.
2. Introduction to Ceramics by: W. D. Kingery, H. K. Brown and D. R. Uhlmann, Wiley Interscience Publication, John Wiley & Sons.
3. Rahaman, M.N., (2017), Ceramic Processing, 2nd Edition, CRC Press

References

1. Tadmor, Z. and Gogos, C.G., (2006), Principles of Polymer Processing, 2nd Edition, Wiley-Interscience.
2. Kalpakjian, S. and Schmid, S.R., (2014), Manufacturing Engineering and Technology, 7th Edition, Pearson.
3. Reed, J.S., (1995), Principles of Ceramics Processing, 2nd Edition, Wiley.
4. German, R.M., (2014), Sintering: From Empirical Observations to Scientific Principles, Butterworth-Heinemann.

Online Course Material

1. Macosko, C.W., Polymer Processing, MIT OpenCourseWare, https://ocw.mit.edu.
2. NPTEL: Processing of Polymers and Composites, IIT Bombay, https://nptel.ac.in.
3. NPTEL: Ceramic Engineering, IIT Kharagpur, https://nptel.ac.in.

Title	**Materials Processing and Characterization Lab**	Number	MTL 2XX0
Department	Materials Engineering	L – T – P [C]	0 – 0 – 4 [2]
Offered for	B. Tech. (MT)	Type	Program Compulsory (PC)
Pre-requisite			

Objectives

The objectives of this course are to:

1. Develop practical understanding of processing and manufacturing techniques for metals, polymers, and ceramics.
2. Acquire hands-on experience in materials characterization techniques such as microscopy, diffraction methods, spectroscopic analysis, and thermal analysis techniques to correlate processing, microstructure, and properties.

Learning Outcomes:

After completing this course, students will be able to:

1. Apply material processing and manufacturing techniques to fabricate and evaluate metallic, ceramic, and polymeric components under different processing conditions.
2. Characterize engineering materials using microscopy, diffraction, thermal, and spectroscopic techniques, and interpret the relationship between structure, properties, and performance.

Course Content

A. Laboratory Experiments of Materials Processing and Manufacturing Lab

1. Injection molding of polymers
2. Blade coating of polymers
3. Sol-gel processing of ceramics
4. Sintering of ceramics
5. Melting and Casting of Aluminum Alloys
6. Welding of steels
7. Welding of Aluminum alloys
8. Additive manufacturing of polymers
9. Additive manufacturing of ceramics

B. Laboratory Experiments of Materials Characterisation Lab

1. Quantitative metallogrphic analysis of metals, ceramics, polymers and composites
2. Electropolishing of aluminium alloys
3. Grain refinement studies using polarization microscope
4. Crystal structure determination using X-ray Diffraction analysis
5. Crystallite size measurement using X-ray Diffraction analysis
6. Indexing of selected area diffraction pattern
7. Specific heat and enthalpy calculation using DSC analysis
8. Phase transformation during solidification using thermal analysis
9. FTIR analysis of ceramic materials
10. UV-Visible spectroscopy analysis of thin films

Course Material

For both labs, specific laboratory protocols and handouts will be made available for different experiments.

Reference Books

1. Zhang,S., Li,L. and Kumar,A., (2008), Materials Characterization Techniques, , CRC Press
2. Cullity, B.D., Stock, S.R., (2014), Elements of X-Ray Diffraction, 3rd Edition, Pearson Education Limited

Detailed Course Content of Programme
Elective Courses

Title	**Processing-Microstructure and Properties of Steels**	Number	MTL 4XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objectives

The Instructor will:

1. Describe the processing-structure-property relationship of steels used in engineering applications

Learning Outcomes

The students are expected to have the ability to:

1. Design the processing routes to obtain steel microstructures that will provide customizable properties to the component for specific engineering applications

Contents

Processing of Steels [6 lectures]: Melting and casting, Steel refining by ESR and VIM, and VAR, heat treatment of steels, thermo-mechanical processing of steels, sheet metal forming and welding of steels

Physical Metallurgy of Steels [9 lectures]: Effects of alloying elements on Fe-C phase diagram, structure and transformation kinetics of pearlite, interphase precipitation, divorced pearlite, Martensite-crystal structure, transformation kinetics, morphology, Bainite-Bainite start temperature, upper and lower bainite, bainitic microstructure, Widmanstattaen ferrite, acicular and equiaxed ferrite

Low Carbon Steels [8 lectures]: types of low carbon steels, hot-rolled ferrite-pearlite steels, processing by hot rolling, cold-rolling and annealing, interstitial free steel, HSLA steel, Dual-phase (DP) steel, TRIP steels, applications

Medium and High Carbon Steels [8 lectures]: microalloyed steel-microstructure and mechanical properties, rail steels, patenting-pearlite formation in high strength steel wire, drawing of pearlitic high carbon steel wires, hardening and tempering process, temper embrittlement, tool steels, applications

Stainless Steels [4 lectures]: alloy design for stainless steels, austenitic, ferritic and martensitic stainless steels, precipitation hardened and duplex stainless steels, applications

Surface Hardened Steels [4 lectures]: carburizing, Nitriding, carbo-nitriding, induction hardening, flame hardening

Textbooks

1. Krauss,G., (2005), *Steels: Processing, Structure, and Performance*, 2nd Edition, ASM International

Reference Books

1. Bhadeshia,H.K.D.H, Honycombe,R.W.K, (2017), *Steels: Microstructures and Properties*, 4th Edition, Butterworth-Heinemann

Title	**Failure of Materials and Prevention**	Number	MTL 4XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objectives

1. To introduce methods for defining and evaluating failure of structural materials and components subjected to steady and time dependent multi-axial (3D) stresses/strains.
2. To introduce materials selection strategies to prevent failure in design.

Learning Outcomes

1. To be able to apply failure theories to predict failure of structural materials and components subjected to various loading conditions.
2. To be able to select materials to prevent failure of components.

Contents

Introduction [2 Lectures]: Review of processing-structure-property relation in materials, types of failure

The Design Process and Materials Data [2 lectures]: Design tools and Materials data, Processes of obtaining materials data, Materials databases

Failure Prediction [8 lectures]: Failure Theories, Yield surface and normality, Plastic Stress-Strain Relationships, Griffith's crack theory, Strain energy release rate, Failure modes in polymers and ceramics, Fracture toughness of engineering materials, Use of Fracture Mechanics Principles in Design

Failure Detection via NDT[4 lectures]: Visual inspection, Dye-penetration technique, surface and subsurface crack detection through eddy current, internal crack detection via ultrasonic technique

Application of Failure Theories under Cyclic Loading [6 Lectures]: S-N Diagram, Cumulative Fatigue Damage, Spectrum Loading, Rainflow counting techniques, Local Strain Life Approach, Damage tolerance and fatigue crack growth in design

Application of Failure Theories under High Temperature [6 lectures]: Creep Mechanism Maps, High Temperature Fracture modes, Presentation of Engineering Creep data, Stress-rupture data, Larson-Miller parameter, Design for Creep

Material Selection for Failure Prevention [11 lectures]: Selection criteria for materials, Material Property Charts, property limits and material indices, Materials indices which include shape, Applications of Failure theories under combined loading and application of Granta Edupack for Material Selection, Case studies on applications to life analysis and design

Textbooks

1. Collins, J.A., (1993), Failure of Materials in Mechanical Design, 2 nd Edition, Wiley-Interscience Publication
2. Hertzberg, R.W., (2013), Deformation and Fracture Mechanics of Engineering Materials, 5 th Edition, Wiley
3. Ashby, M.F., (2016), Materials Selection in Mechanical Design, 5 th Edition, Butterworth-Heinemann

Reference Books:

1. Courtney, T. H., (2005), Mechanical Behaviour of Materials, 2 nd Edition, Waveland Press Inc.
2. Jones, D.R.H., and Ashby,M.F., (2011), Engineering Materials 1: An Introduction to Properties, Application and Design, 4 th Edition, Butterworth-Heinemann
3. Shigley, J. E., Mischke, C., Nisbett, K., and Richard, B., (2008), Mechanical Engineering Design, McGraw Hill Education
4. William T. Becker; Roch J. Shipley, Failure Analysis and Prevention, Volume 11, ASM Handbook

Online Course Material

1. Bhattacharya, B., Materials Selection and Design, NPTEL Course Material, Department of Mechanical Engineering, Indian Institute of Technology Kanpur, http://nptel.ac.in/courses/112104122/
2. Sen, I., Fracture, Fatigue and Failure of Materials, NPTEL Course Material, Department of Metallurgical and Materials Engineering, Indian Institute of Technology Kharagpur, https://onlinecourses.nptel.ac.in/noc22_mm42/

Title	**Composites**	Number	MTL 4XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objectives

The main objective of this course is

1. To introduce structure and processing of composites
2. To and uses of composites

Learning Outcomes

The students are expected to

1. Classification and application of different composite materials and their processing techniques
2. Acquire understanding about the structure property relationship of different types of composites

Contents

Introduction and Classification of Composites [6 lectures]: Composites, matrix*:* metal matrix composites, polymer matrix composites, ceramic matrix composites, fiber reinforcements: fiber types, particulate reinforcements, nanocomposites

Design and Manufacturing of Composites [11 lectures]: Selection of matrix and reinforcement- Reinforcement with Aligned Continuous Fibers and Discontinuous fibers, wettability and role of interface; Manufacturing- Hand and Prepeg layup, Bag molding, Autoclave Processing, Compression Molding, Resin Transfer Molding, Pultrusion; Metal-Matrix Composites- Solid state and Liquid State processing; Ceramic-Matrix Composites- Solid state, sol-gel, co-precipitation and hydrothermal techniques

Tailoring Composite Materials [6 lectures]: by component selection, by interface modification, surface modification, microstructure control

Basic Mechanics of Composites [8 lectures]: Fiber-Matrix interface- theories of adhesion, glass fiber-polyester resin interface, measurement of bond strength, Elastic properties-unidirectional lamiae, elastic properties of short fiber composite materials, strength of a unidirectional laminae, fiber pull out, longitudinal and transverse tensile and compressive strength, in-plane shear strength

Properties and Applications of Composites [8 lectures]: Monotonic properties, fatigue and creep properties of composites, *Applications of Composites*- Materials for Lightweight Structures, Materials for civil infrastructure, Materials for Aerospace components- propeller blades, turbine blades, space applications

Textbook

1. Clyne, T. W. and Hull, D., (2019) *An Introduction to Composite Materials*, 3rd edition, Cambridge University Press.
2. Balasubramanian, M., (2013), Composite Materials and Processing, 1st Edition, CRC Press

Reference Book

1. Matthews F. L. and Rawlings R. D. (1999) *Composite Materials*, 1st edition, Woodhead Publishing.

Online Course Material

1. Prof. Nachiketa Tewari, Introduction To Composites, Aerospace Engineering, Indian Institute of Technology Kanpur, https://nptel.ac.in/courses/112/104/112104229/

Title	**Heat-Treatment of Metals and Alloys**	Number	MTL 4XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objectives

1. To understand the foundation of heat treatment processes affecting the microstructures.
2. To learn different types of furnaces used in heat treatments and heating mechanisms.
3. Heat transfer mechanisms in heat treatment processes and associated microstructures.

Learning Outcomes

1. Concept behind the microstructure under equilibrium and non-equilibrium cooling. To acquaint with TTT and CCT diagrams and associated microstructure under different industrial cooling conditions.
2. Significant roles of microstructural features on mechanical properties of metals and alloys.

Contents

Heat treatment process variables; Heating and cooling of steels for heat treatment; Formation of austenite, Transformation of austenite to pearlite, bainite and martensite; Characteristics of transformation products; Homogeneity of austenite; Austenite grain size, Determination and importance of austenitic grain size [7 Lectures]

T-T-T-and C-C-T diagrams: Factors affecting T-T-T curves; Prediction of T-T-T-and C-C-T diagrams; Heat treatment processes: different types of annealing, spheroidizing, normalzing, hardening, tempering, patenting, austempering, martempering, sub-zero treatment, Application of Thermo-Calc and DICTRA [9 Lectures]

Thermo-mechanical treatment of steels: Ausforming, Isoforming, Cryoforming; Heat removal mechanisms; Hardenability of steels; Heat treatment defects, Residual stress upon heat treatment, Industrial heat treatment practices [6 Lectures]

Surface heat treatments - Carburizing, Nitriding, Cyniding and Carbonitriding, Flame hardening, Induction hardening, Laser hardening [4 Lectures]

Heat treatment of some steels: IF steel, DP steel, TRIP steel, high speed steels, maraging steels, spring steels. [4 Lectures]

Age Hardening; Heat treatment of non-ferrous metals and alloys - Aluminium alloys, Magnesium alloys, Copper alloys, Titanium alloys and Nickel alloys. [7 Lectures]

Practical considerations in heat treatment: Industrial Heat Treatment Processes, Types of furnaces and furnace atmospheres, Cooling media [2 Lectures]

Text Books

1. Rajan TV, Sharma, CP and Sharma, A. (2011), Heat Treatment: Principles and Techniques, 2 nd edition, Prentice Hall India.
2.Krauss G. (1990), Steels - Heat Treatment and Processing Principles, ASM International, Materials Park, Ohio.
3. Bhadeshia,H.K.D.H, Honycombe,R.W.K, (2017), Steels: Microstructures and Properties, 4 th Edition, Butterworth-Heinemann

Reference Books

1. Thelning KE (1984), Steel and its Heat treatment, 2 nd edition, Butterworth, London.
2. ASM, Metals Hand Book: Heat Treating (1990), Vol. 4, 9th Ed., Metals Parks, Ohio.

Title	**Solidification of Metals and Alloys**	Number	MTL 7XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objective:

1. The fundamentals of liquid to solid transformation and practical implication to practice.
2. Processes such as crystal growth, interface stability, splat cooling, ingot casting shape casting will be dealt with paying particular emphasis to fundamentals of mass transport, heat flow and interface kinetics.

Learning Outcomes

1. Build up knowledge on fundamentals of solidification of metals and alloys and correlating with various casting processes and microstructure evolution.

Contents

Fundamentals of solidification [8 Lectures]: Thermodynamics of nucleation and growth phenomena, crystal growth, homogenous and heterogeneous nucleation, role of contact angle grain refinement.
Solidification of metals and alloys [9 Lectures]: Heat flow, plain front solidification, cellular solidification, constitutional undercooling, formation of dendrites, eutectic solidification, peritectic solidification, monotectic solidification.
Solidification processes *[8 Lectures]*: Solidification during welding, additive manufacturing, zone refining, directional solidification, single crystal fabrication, continuous casting.
Rapid solidification processes (RSP) [6 Lectures]: Conventional and unconventional effects. Undercooling and recalescence. Amorphous state. Glass forming ability. Methods for preparing rapidly solidified material.
Recent developments [8 Lectures]: interface kinetics, interface dynamics, phase selection, microstructure selection. Effect of vibration. Theory convection effects. Solidification under external field, electromagnetic stirring, ultrasonic processing. Multicomponent alloys and numerical techniques.

Text Books

1. Flemings, M.C., *Solidification Processing*, Mcgraw-Hill Book Company, 1974.
2. Campbell, J., *Complete casting Handbook: Metal Casting Processes, Metallurgy, Techniques and Design*, Butterworth-Heinmann, 2015.

Reference Books

Minkoff, I., *Solidification and Cast Structure*, Wiley, 1986.

Title	**Light Metals & alloys**	Number	MTL 7XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objective

1. Introduce the prospect of competitive structural materials with high specific strength.

Learning Outcomes

1. Appreciate the role of light metals for energy savings in different applications/sectors.

Contents

Introduction (4 lectures): Importance of light metals, overview of light metals production.
Techniques for melting and solidification (6 lectures): solidification, grain refinement, casting processes.
Heat treatment (7 lectures): grain refinement, strengthening by solid solution, precipitation hardening, dispersion of second phase particles.
Alloy designations and properties (7 lectures): specific alloy systems ((a) Al-alloys, (b) Mg-alloys, (c) Ti-alloys).
Manufacturing and applications (7 lectures): aerospace, biomedical, automobile, domestic appliances.
Novel processing methods (8 lectures): composites, metallic foams, rapid solidification, Quasi crystals, amorphous alloys, mechanical alloying, physical vapor deposition, additive manufacturing.

Text Books

1. Polmear, I. J., St. John, D., Nie, J. F., Qian, M, *Light Alloys*, 5th edition, Elsevier 2016.
2. Brandes E. A. and Brook G. B., *Smithells Light Metals Handbook*, Elsevier, 1998.

Self-Learning materials

1. Totten G.E. and Mackenzie D.S., *Handbook of Aluminum Vol. 1: Physical Metallurgy and Processes*, CRC Press 2003.
2. Friedrich H.E., and Mordike B.L., *Magnesium Technology*, Springer, 2004.

Lütjering G., Williams J.C., *Titanium*, 2nd edition, Springer, 2007.

Title	**High Temperature Materials**	Number	MTL 7XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objectives

1. To understand the structure, properties and behavior of materials on elevated temperature.
2. To gain knowledge in the selection of suitable materials for various engineering applications at high temperature.

Learning Outcomes

1. Enrich knowledge of various behavior and property changes inside the material structure in raised temperature and methods to strengthening the material.

Contents

Creep (8 Lectures): Factors influencing functional life of components at elevated temperatures, definition of creep curve, various stages of creep, metallurgical factors influencing various stages, effect of stress, temperature and strain rate.

Design for creep resistance (8 lectures): Design of transient creep time, hardening, strain hardening, expressions of rupture life of creep, ductile and brittle materials, Monkman-Grant relationship.

Fatigue and fracture (5 lectures): S.N. curves - Endurance limits - Effect of mean stress, Goodman, Gerber and Soderberg relations and diagrams, Various types of fracture, brittle to ductile from low temperature to high temperature, cleavage fracture, and ductile fracture due to micro void coalescence-diffusion controlled void growth; fracture maps for different alloys and oxides.

Oxidation and hot corrosion (8 lectures): Oxidation, Pilling, Bedworth ratio, kinetic laws of oxidation- defect structure and control of oxidation by alloy additions, hot gas corrosion deposit, modified hot gas corrosion, fluxing mechanisms, effect of alloying elements on hot corrosion, interaction of hot corrosion and creep, methods of combat hot corrosion.

Super alloys and other materials (10 lectures): Iron base, Nickel base and Cobalt base super alloys, composition control, solid solution strengthening, precipitation hardening by gamma prime, grain boundary strengthening, TCP phase, embrittlement, solidification of single crystals, Intermetallics, High temperature ceramics-Smart materials

Textbook

1. Raj. R., *Flow and Fracture at Elevated Temperatures*, American Society for Metals, USA, 1985.
2. Hertzberg R. W., *Deformation and Fracture Mechanics of Engineering materials*, 4th Edition, John Wiley, USA, 1996.
3.Courtney T.H, *Mechanical Behavior of Materials*, McGraw-Hill, USA, 1990.

Reference Books

1. Boyle J.T, Spencer J, *Stress Analysis for Creep*, Butterworths, UK, 1983.
2. Bressers. J., *Creep and Fatigue in High Temperature Alloys*, Applied Science, 1981.
McLean D., *Directionally Solidified Materials for High Temperature Service*, the Metals Society, USA, 1985.

Title	**Battery Materials**	Number	MTL 4XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objectives

Through this course the instructor

1. Will provide an insight of development of battery materials

Learning Outcomes

The students are expected to gain knowledge on:

1. Cycling behaviour of the batteries.
2. Development of different types of batteries demanding on application, Merits and demerits of each batteries.
3. Materials developed so far and future goals to be achieved.

Contents

Introduction to Electrochemistry [8 lectures]: Chemical and Electrochemical Reactions, Components of electrochemical cells, Operation of a Cell, Self-Discharge, Theoretical Cell Voltage, Capacity, and Energy, Specific Energy and Energy Density, Thermodynamic Background, Variation of the Voltage with Discharging and Recharging, Cycling Behaviour.

Battery Characteristics, Performance, Design, Selection & Standardization [2 lectures]: General Characteristics and Factors Affecting Battery Performance, Battery Design and Construction, Selection and Application of Batteries, Battery Standardization

History and evolution of Batteries [6 lectures]: Primary Batteries, voltaic pile, Daniell cell, Zinc-Carbon Batteries, Alkaline-Manganese Dioxide Batteries, Secondary Batteries, Lead-Acid Batteries, Nickel-Iron Batteries, Nickel- Cadmium Batteries, Nickel-Zinc Batteries, Nickel-Hydrogen Batteries, Nickel-Metal Hydride Batteries, Lithium-Ion Batteries.

Development of Negative Electrodes [8 lectures]: The lead electrode, Aluminium and Magnesium electrode, Zinc and Cadmium Electrode in Aqueous Systems, Metal Hydride Electrodes, Negative Electrodes in Lithium Cells

Development of Positive Electrodes [8 lectures]: Manganese Dioxide/ The "Nickel" Electrode / Lead oxide electrode/ Oxygen electrode Electrodes in Aqueous Systems.

Development of Electrolytes [7 lectures]: Liquid Electrolytes, General Considerations Regarding the Stability of Electrolytes Vs. Alkali Metals, Elevated Temperature Electrolytes for Alkali Metals, Electrolyte Stability Windows and Their Extension, Composite Structures That Combine Stability Regimes, Solid Electrolytes.

Textbook

1. Huggins R. A. (2009) *Advanced Batteries: Materials Science Aspects*, 1st Edition, Springer

Reference Books

2. Linden D. and Reddy T. B. (2002) *Handbook of Batteries, 3rd Edition*, The McGraw-Hill Companies

Online Course Material

Prof. Bazant M., MITopencourseware: Electrochemical Energy Systems, MIT Course Number 10.626 / 10.426, https://ocw.mit.edu/courses/chemical-engineering/10-626-electrochemical-energy-systems-spring-2014/

Title	**Functional Ceramics**	Number	MTL 4XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objectives

The instructor will:

1. Provide in-depth knowledge into the electronic, magnetic, and optical properties of ceramic materials systems and their properties

Learning Outcomes

On successful completion of the course, the students would be:

a) Know the relationships between ceramic structure (electronic, microstructural, and defect structure) and the wide variety of properties they display.

b) Learn how these variety of phenomena can be exploited in useful applications like sensors, and piezoelectric transducers

c) Access Current day research topics on electronic magnetic and optical materials.

Contents

General Introduction [1 Lecture]

Crystallography and Bonding [4 Lectures]: Structures, Bonding in Ceramics Structure formation rules in Oxides, Important structures in Oxides.

Defects [5 Lecture hours]: Defects in ceramics and defect chemistry, Defects diffusion and equilibrium

Electrical conductivity [5 Lectures]: Conduction Mechanisms in Ceramics, Resistors, varistors and thermistors

Dielectrics, ferro- and piezo-electrics [15 Lectures]: Capacitor Materials, Dielectric Breakdown, High and low permittivity materials, Microwave and tunable dielectrics, Ferroelectric memories, Piezoelectric materials: PZT, $BaTiO_3$, Morphotropic Phase boundary, Electrostriction, Pyroelectric materials: Thermal Imaging, IR detection

Magnetic Ceramics [3 Lectures]: General Properties, Magnetic ferrites, Magneto crystalline anisotropy, Magnetostriction, Applications

Electro-optical Properties [3 Lectures]: Electro-optic effect, Non-Linear optics, Birefringence, PLZT material and structure

Material processing [3 Lectures]: Thin film and Bulk processing

Textbook

1. Electroceramics: Materials, Properties and Applications by A. J. Moulson and J. M. Herbert, Chapman and Hall, 2nd Edition, 1990.
2. Kofstad, Per. "Nonstoichiometry, diffusion, and electrical conductivity in binary metal oxides." (1972).

Reference Books

1. Introduction to solid State Physics by Charles Kittel, Wiley Eastern, 5th Edition, 1983
2. Newnham, R. E. (2012). Structure-property relations (Vol. 2). Springer Science & Business Media

Online Course Material

1. Ashish Garg, Electroceramics, NPTEL Course Material, Department of Material Science & Engineering, Indian Institute of Technology Kanpur, https://nptel.ac.in/courses/113/104/113104005/
2. Ashish Garg, Fundamentals and Applications of Dielectric Ceramics, NPTEL Course Material, Department of Material Science & Engineering, Indian Institute of Technology Kanpur, https://nptel.ac.in/courses/113/104/113104090/

Title	**Magnetism and Magnetic Materials**	Number	MTL 4XX0
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objectives

The course will provide knowledge of underlying phenomena of magnetism. Emphasis will be on the preparation process & characterisation methods of magnetic materials used for Engineering applications

Learning Outcomes

The students are expected to have the ability of understanding magnetic phenomena, and metallurgical aspect of developing magnetic materials. They will also get an exposure of engineering application of magnetic methods and magnetic materials.

Contents

o Origin of magnetism, Bohr Magnetron, Classification of magnetic materials, Weiss molecular field (5L)
o Magnetism in Metals and Insulators, Exchange Interaction, Magnetic Anisotropy and Magnetostriction, (6L)
o Domain and Domain Walls, Magnetic Hysteresis and Barkhausen Emissions, Techniques for Characterisation of Magnetic Materials, Superconductor and Superconducting Magnets, Structure Property Relations (8L),
o Soft Magnetic Materials: Nickel Iron Alloys, Cobalt-Iron alloys, Electrical steel (CRNO, CRGO and amorphous), Amorphous & Nanostructured magnetic alloys. (6L)
o Hard Magnetic Materials: Alnico, NdFeB, SmCo (6L)
o Soft & Hard Ferrites, Magnetic Shape Memory and Magnetocaloric Materials (6L),
o Application of Magnetism and Magnetic Materials (2L)

Text/Reference Book:

(i) Chih-Wen Chen, Magnetism and Metallurgy of Soft magnetic Materials, Dover Publications, INC. New York, ISBN 0-486-64997
(ii) B.D. Cullity, Introduction to Magnetic Materials, Addison-Wesley publishing company, California, London,
(iii) D. C. Jiles, Introduction to Magnetism and Magnetic Materials, Chapman & Hill, London, 1991.
(iv) S. Chikazumi, Physics of Ferromagnetism, Oxford University Press 1997, Reprinted in 2002, ISBN: 0-19-851776-9

Title	**Solar Cell Materials and Devices**	Number	MTL 7760
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objectives

The course provides an understanding of the solar cell materials with the latest advancements, focusing on their properties, fabrication, characterization, and applications.

Learning Outcomes

The students are expected to **(i)** understand the fundamental properties of materials used in solar cells. **(ii)** Learn the solar cell fabrication processes and performance factors **(iii)** Analyze the structural, electrical, and optical properties of solar cell materials using relevant characterization techniques.

Contents

Introduction [4 lectures]: Photovoltaic (PV) Technology, Electromagnetic Radiation and Energy Harvesting, Energy Conversion Mechanisms in Solar Cells, Fundamental Limits to Photovoltaic Conversion.

Basics of Solar Cells [4 lectures]: Physics of p-n junction diode under dark and illumination, Solar Cell Equivalent Circuit, Characteristics of Solar Cells (V_{OC}, J_{SC}, FF, Efficiency, Series and Shunt Resistance), Types of Solar Cells, Light management in Solar cells.

Silicon based Solar Cells [6 lectures]: Introduction to Silicon Solar Cells, Monocrystalline Silicon Solar Cells, Polycrystalline Silicon Solar Cells, Amorphous Silicon Solar Cells, Fabrication Methods, Scope of Improvement.

III-V Solar Cells: Materials and Applications [5 lectures]: Homo and Hetero junction III-V Solar Cells, GaAs Solar Cells, InP Solar Cells, InGaAsP Solar Cells, GaN, Multijunction Solar Cells, Fabrication Methods.

Thin Film Solar Cells: [6 lectures]: Introduction and Advantages, Thin Film Solar Cell Materials - Binary and Quaternary Chalcogenides, Fabrication Methods.

Emerging Solar Cells [8 lectures]: Dye-sensitized solar cells, Perovskite solar cells, Organic Solar Cells, Hybrid Solar Cells, Quantum Dot and Nanostructured Solar Cells, Tandem Solar Cells, Bifacial Solar Cells.

Characterization and Simulation Studies [6 lectures]: Structural and Morphological Characterization, Optical (UV-Vis, PL and TRPL) and Electrical Characterization: Hall Effect and I-V Measurements, Case Studies on Simulation-Driven Performance Optimization of Solar Cells.

Textbooks:

1. Thomas Dittrich, Materials Concepts for Solar Cells, Imperial College Press, 2014.
2. Jenny Nelson, The Physics of Solar Cells, Imperial College Press, 2003
3. S.K. Sharma, K. Ali, Solar Cell from materials to device technology, Springer Nature, 2020.
4. J. Poortmans, and V. Arkhipov, Thin Film Solar Cells: Fabrication, Characterization, and Applications, Wiley, 2006.

Reference Books:

1. Laurentiu Fara, Masafumi Yamaguchi-Advanced Solar Cell Materials, Technology, Modeling, and Simulation, Engineering Science References, 2013.
2. Peter Würfel, Physics of Solar Cells: From Basic Principles to Advanced Concepts, Wiley-VCH, 2016.

Video link: NPTEL

NPTEL, Solar Photovoltaics: Principles Technologies and Materials
https://www.youtube.com/watch?v=Xcy6rTYxijQ
NPTEL-NOC IITM
(https://www.youtube.com/watch?v=tSD8lgdicNk)
NPTEL, Electronic materials, devices, and fabrication
https://www.youtube.com/watch?v=mCgXsEyQZSI

Title	**Advanced Semiconductor Materials and Device Applications**	Number	MTL 7550
Department	Materials Engineering	L – T – P [C]	2 – 0 – 2 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objectives

1. To develop an understanding of the layered and non-layered semiconductor materials synthesis/growth and characterization techniques used in the semiconductor industry.

Learning Outcomes

The students will be able to:

1. Understand the basics of advanced semiconductor materials properties, failure analysis methods, reliability, and characterization.
2. Fabricate semiconductor layered material devices and analyse their properties.

Contents

Fractal-1: Material Synthesis and Characterization

Introduction, Basics of semiconductor, Crystallographic properties of the semiconductors. *(7 lectures)*

Growth and synthesis of semiconductor materials (2D and 3D), Defects and its characterization and Advantages of 2D materials. *(6 lectures)*

Fractal-2: Device Characterization

Electrical characterization of semiconductors: Material influenced test structure based electrical characterization methods like: two-probe/four-probe, Van der Pauw method, Hall-effect measurements etc. *(5 lectures)*

Characterization of junctions and diodes: C-V measurements, Life-time measurement, Metal-semiconductor: Schottky contacts, Barrier-height measurement, I-t techniques. *(4 lectures)*

Reliability and failure analysis: Gate oxide reliability measurement, Time-to-breakdown and charge to breakdown and Failure analysis, Noise measurements and analysis. *(4 lectures)*

Lab:

1. Synthesis of semiconductor materials atomic sheets of Graphene Oxide (MT).
2. Synthesis of semiconductor elemental sheets of Xenes and Transition metal dichalcogenides (TMDCs) (MT).
3. Fabrication of p-n junction (Graphene/Si/SiO_2, G/MoS_2, etc.) (MT).
4. Characterization of p-n junction (Graphene/Si/SiO_2, G/MoS_2, etc.) (MT).
5. Electrical characterization of the fabricated devices (EE).
6. Reliability and Failure analysis (EE).
7. Material process (spectroscopic) and Quality control (MT).
8. Tailoring the electrical, optical and Youngs' modulus of the semiconducting materials (MT).
9. Application of semiconductor in sensing (SERS/gas/strain) (MT/EE).
10. Application of semiconductor in opto-electronics (MT/EE).

Textbook

1. Schroder, D.K., *Semiconductor Material and Device Characterization,* 3rd edition, Wiley-Interscience, New York, 2006.
2. Nicollian, E.H., Brews, J., *MOS Physics and Technology,* ISBN: 978-0-471-43079-7, 1982.

Online Course Material

1. Haridoss, P., *Physics of Materials*, NPTEL Course Material, Department of Metallurgy & Material Science, Indian Institute of Technology Madras, https://nptel.ac.in/courses/113/106/113106039/.

Title	**Functional 2D Materials**	Number	MTL 7540
Department	Materials Engineering	L – T – P [C]	2 – 0 – 2 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objectives

1. To introduce the emergent field of functional 2D materials.
2. Provides a comprehensive overview of 2D materials their hybrids and heterostructures systems that are relevant to different scientific applications.

Learning Outcomes

Upon completion, students are expected to

1. To know the basics of 2D materials.
2. Van der Waal and Non- van der Waal materials hybrids and heterostructures.
3. Design and fabricate 2D Advanced Materials based devices.

Contents

Fundamentals of 2D Materials

Introduction of functional quantum materials, 2D family, Electronic, Magnetic, Optical properties of functional quantum materials, Dirac metals, Weyl fermions. *(5 lectures)*

2D functional quantum materials at atomic scale, Topological insulators, Computation of electronic properties of functional quantum materials, Investigation of defect and Ohmic/Schottky contacts. *(4 lectures)*

Fundamental and properties of two-dimensional materials in sensing: Introduction, Band Alignment, Parameters of sensor performance. *(4 lectures)*

Fabrication of Devices, Application and Challenges

Synthesis (top-down and bottom-up approach and challenges), Characterization, and transfer of Van der Waals (graphene, boron nitride, Molybdenum disulphide, Tungsten Disulphide, Black Phosphorus, etc.) atomic sheets, Microscopic characterization by TEM. *(3 lectures)*

Non-Van der Waals Materials (Borophene, Hematene, Plumbene, MXene, etc.) their Hybrids (2D/1D, 2D/2D, 2D/3D), composites and heterostructures, Fabrication of 2D Material devices and challenges. *(3 lectures)*

Application of 2D Materials and their hybrids as well as heterostructures: Introduction (Graphene, Xenes, MXene, and Complex 2D Materials) Opto-Electronics (LEDs), Gas Sensor, Bio-Sensor, Drug-Delivery, Filtration/Desalination, Photovoltaics, Energy Storage and Generation, Lubricants, Environmental Remediation. *(5 lectures)*

Challenges with 2D Materials hybrids/heterostructures and device fabrications. *(2 lectures)*

Lab:

1) Synthesis of Van der Waals 2D material (e.g. Graphene, TMDCs) atomic sheets.
2) Characterization (morphological and spectroscopic) of the atomic sheets.
3) Fabrication of nano-devices (bio sensors, gas sensors, etc.).
4) Failure analysis and Quality control.
5) Synthesis of non-Van der Waals 2D material (e.g. Hematene, borophene, etc.) atomic sheets.
6) Characterization (morphological and spectroscopic) of the atomic sheets.
7) Fabrication of nano-devices (bio sensors, gas sensors, etc.).
8) Failure analysis and Quality control.
9) Synthesis of 2D hybrids and heterostructres (e.g. graphene/graphene oxide/reduced graphene oxide/borophene/hematene/TMDCs, etc.) atomic sheets.
10) Characterization (morphological and spectroscopic) of the atomic sheets.
11) Failure analysis and Quality control.

Textbook

1. Zettilli, N., Gubernatis, J.E., Lookman, T., *Quantum Mechanics: Concepts and Applications, Wiley*, 2009.

2. Griffiths, D.J., *Introduction to quantum mechanics*, Prentice Hall International, 2004.
3. Banks, C., E., Browson, D.A.C., *2D Materials, Characterization, Production and Applications*, CRC Press, 2017.

Reference Books

1. Ng, L.W.T., Hu, G., Howe, R.C.T., Zhu, X., Yang, Z., Jones, C., Hasan, T., *Printing of Graphene and Related 2D Materials: Technology, Formulation and Applications*, Springer, 2019.
2. Errana, G., *Metal Oxide Nanostructures as Gas Sensing Devices,* CRC Press, 2012.
3. Avouris, P., Heinz, T.F., Low, T*., 2D Materials Properties and Devices*, Cambridge University Press, 2017.

References

1. Paglione, J., Butch, N.P., Rodriguez, E. E., *Fundamentals of Quantum Materials: A Practical Guide to Synthesis and Exploration,* World Scientific Pub Co Inc, 2021.
2. Heitmann, D., Quantum Materials*, Lateral Semiconductor Nanostructures, Hybrid Systems and Nanocrystals,* Springer, 2010.
3. Bartolomeo, D., *2D Materials and Van der Waals Heterostructures,* MDPI, ISBN 978-3-03928-769-7, https://doi.org/10.3390/books978-3-03928-769-7.
4. Hywel, M., Rout, C.S., Late, D.J., *Late Fundamentals and Sensing Applications of 2D Materials,* Elsevier, 2021.

Title	**Functional Materials**	Number	MTL 7150
Department	Materials Engineering	L – T – P [C]	2 – 0 – 2 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objectives

1. To provide the students with a detailed understanding (synthesis/growth, structure, and properties) of the functional materials.
2. To develop an understanding of the usage of functional materials as a component of modern devices.

Learning Outcomes

The students will be able to:

1. Compare, correlate and investigate the different synthesis, characterization, and application of exemplar functional materials.
2. Integrate the understanding of functional materials' properties and their applications.

Contents

Material synthesis and processing of functional materials: Bulk Synthesis: Solid state synthesis, Sol-Gel synthesis, Thin-Film synthesis: Sputtering, Molecular beam epitaxy, CVD, 3D printing, electrospinning. *(7 lectures)*

Defects in solids: Defect types and dimensionality effect on defects, Characterization (morphological and spectroscopic), Control of defects. *(7 lectures)*

Semiconductor materials: Band structure, Doping, Band-Gap engineering. *(2 lectures)*

Applications: Beyond Si semiconductors, GaN, GaAs, SiC, Ga_2O_3, LEDs, and photovoltaic cells (CdS, CIGS, CZTS, Perovskites and Organic solar cell materials). *(6 lectures)*

Materials for energy applications: Thermoelectric materials: ZT value, Band-Gap, Conductivity engineering, Oxide materials, Heusler alloys, Artificial and hierarchical materials, Peltier cooling, Thermoelectric generator, , Dielectric, Piezoelectric, Ferroelectric materials and applications, *(7 lectures)*

Magnetic materials & applications: Magnetic exchange energy, anisotropy energy, Magnetic domains, Application of soft and hard magnetic materials, Magnetic data storage, Superconductors in electronics. *(6 lectures)*

Optical materials: Optical lithography, and applications, Electro-optic materials. *(4 lectures)*

Textbook

1. Askeland, D.R., Phule, P.P., Wright, W.J., *The Science and Engineering of Materials*, 6th edition, Cengage Learning, 2010.
2. Callister, W.D., Rethwisch, D.G., *Materials science and Engineering: An Introduction*, 8th edition, Wiley, 2010.
3. Mitchell, B.S., *An Introduction to Materials Engineering and Science for Chemical and Materials Engineers*, 1st ed., Wiley- Interscience, 2003.
4. Kittel, C., *Introduction to Solid State Physics*, 8th edition, Wiley, 2005.

Reference Books

1. Kasap, S.O., *Principles of Electronic Materials and Devices*, 3rd edition, McGraw-Hill, 2006.
2. Raghavan, V., *Materials Science & Engineering: A first course*, 5th edition, PHI Learning, 2004.

Online Course Material

1. Haridoss, P., *Physics of Materials*, NPTEL Course Material, Department of Metallurgy & Material Science, Indian Institute of Technology Madras,
https://nptel.ac.in/courses/113/106/113106039/.
2. Garg, A., *Electroceramics*, NPTEL Course Material, Department of Material Science & Engineering, Indian Institute of Technology Kanpur,
https://nptel.ac.in/courses/113/104/113104005/.

Title	**Computational Thermodynamics**	Number	MTL 4XXX
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objectives

1. Understanding the correlation between the phase diagram and chemical thermodynamics for alloy design.

Learning Outcomes

1. Ability to construct multicomponent phase diagrams using computational thermodynamic approach.

Contents

Thermodynamics of Solution [7 Lectures]: Thermochemistry of solutions. Evolution of phase diagrams - quasichemical theory, Sub regular solution model.

Calculation of thermodynamic parameters [8 Lectures]: First principle calculation, Semi-empirical- Miedema approach. Calorimetric measurement, electrochemical measurement,

Calculation of multicomponent phase diagram [14 Lectures]: Redlitch-kister polynomial, Muggianu and kohler extrapolation. Crystallography in thermodynamics, Order and disorder structure, antisite defect and vacancies. Compound energy formalism. Modeling of Interstitial and substitutional phases, stoichiometric and non-stoichiometric compounds.

How to set-up simulation/calculation [10 Lectures]: a) Construction of equilibrium and non-equilibrium phase diagram, b) Vertical section phase diagram and Property diagram for multi-component alloys, c) Scheil and equilibrium solidification simulation, d) Liquidus projections and monovariant reactions in ternary phase equilibria, e) Creation of user defined thermodynamic database.

Textbooks

1. Saunders and Miodownik, (1998), *CALPHAD (Calculation of Phase Diagrams): A Comprehensive Guide,* 1st Edition, Pergamman Press,
2. Lukas,H., Suzana,G., Fries, B.S., (2007), *Computational Thermodynamics: The Calphad Method,* Cambridge University Press,

Self-Learning Material

1. Porter, D.A., Easterling, K. E., and Sherif, M.Y., (2009), *Phase Transformation in Metals and Alloys*, 3rd Edition, CRC Press

Online Course Materials

1. Murty, B.S., Advanced Metallurgical Thermodynamics, NPTEL Course Material, Department of Metallurgical and Materials Engineering, Indian Institute of Technology Madras, https://nptel.ac.in/courses/113106031/.

Title	**Introduction to Dislocation Dynamics**	Number	MTL 4XXX
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objective

1. Introduce basic concepts of discrete dislocation dynamics simulations

Learning Outcomes

The student will be able to

1. Create a simulated system with various dislocation configurations.
2. Determine the properties associated with the dislocations by running simulations.

Course Content

1. Crystal dislocations, The burgers vector, Motion of a crystal dislocation, Atomistic mechanism of dislocation motion, introduction to atomistic simulations (6 lectures)
2. Static simulations: setting up initial configuration, boundary conditions, data analysis and visualisation: Energy filtering and centro-symmetry deviation (6 lectures)
3. Dynamic simulations: Creation of dislocation through misfit at interfaces, stress control and temperature control, extraction of dislocation velocity (6 lectures)
4. Periodic boundary conditions: Displacement field induced by a dislocation, dislocation core energy, Peierls stress (6 lectures)
5. Free energy calculations: harmonic approximation, a Vacancy in the Dislocation Core, beyond harmonic approximation (4 lectures)

Hands-on (Computer simulations) (11 lectures):

1) An atomistic simulation of an edge and a screw dislocation in BCC tantalum
2) Simulation of a system containing vacancy–interstitial pair and data analysis
3) Simulation of a perfect grain boundary and data analysis
4) Simulation of dislocation glide motion at different stress values
5) Simulation a system with a screw dislocation dipole and estimation of the excess energy associated with the dislocation dipole
6) Estimation of dislocation core energies
7) Estimation of a Peierls stress near dislocations
8) Compute the free energy of a silicon crystal containing a vacancy

Text Books

1. Bulatov, Vasily, and Cai, Wei. Computer Simulations of Dislocations. United Kingdom, OUP Oxford, 2006.

References

1. Cui, Yinan. The Investigation of Plastic Behavior by Discrete Dislocation Dynamics for Single Crystal Pillar at Submicron Scale. Singapore: Springer Nature Singapore, 2016.

Title	**Machine Learning for Materials Design**	Number	MTL7520
Department	Metallurgical and Materials Engineering	L–T–P: C	2–0–2 [3]
Offered for	M.Tech.	Type	Elective
Pre-requisite	-		

Objectives

1. Introduce a selected set of Machine learning and Deep Learning techniques for Materials design.
2. Introduce hands-on experience with some practical methods and techniques in materials informatics for materials classification, regression, clustering, and adaptive design.

3. Introduce hands-on experience in analyzing data and applying ML approaches through a set of case studies pertaining to Materials Engineering.

Learning Outcomes

1. Aware of different data-driven techniques for creating Materials Knowledge.
2. To use machine learning tools for solving problems in materials design

Contents

Deep Learning Techniques:
Supervised Learning using deep networks: Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNNs), Self-Supervised Learning, Few Shot Learning, adversarial Learning, and Transformers. (13 lectures)

Unsupervised Learning using deep networks: Auto encoders, Deep Belief Networks (5 lectures)

Ensemble Classifiers: Random Forests, Gradient Boosted Decision Trees (4 lectures)

Association Analysis: Mining association rules, Graph Neural Networks (4 lectures)

Laboratory exercises:

1. A deep-learning technique for phase identification using synthetic XRD powder patterns
2. Physics-driven ML for catalysts design
3. Data-Driven Prediction of Interatomic Potentials
4. Automated Micrograph Analysis: Optical, SEM, TEM and EBSD analysis.
5. Data-Driven Approach for design and rapid screening of High Entropy and Amorphous alloys
6. Materials Property Predictions Based on Data from Chemistry, Structure and Processing
7. Data Driven Modelling for Defect prediction during Additive Manufacturing Process
8. Fatigue life prediction using ML

Books

1. Pilania, G., Balachandran, P. V., Gubernatis, J. E., and Lookman, T., *Data-Based Methods for Materials Design and Discovery: Basic Ideas and General Methods*, Morgan & Claypool Publishers, 2020.

2. *Machine Learning And Data Mining In Materials Science*, Edited by Huber, N., Kalidindi, S. R., Klusemann, B. and Cyron, C. J., Frontiers in Materials, 2020.

3. Mueller, T., Kusne, A.G. and Ramprasad, R., *Machine Learning in Materials Science.* In Reviews in Computational Chemistry, Edited by Parrill, A.L., and Lipkowitz, K.B., John Wiley & Sons, 2016.

Reference Books:

1. Materials Informatics: Methods, Tools, and Applications, Edited by Olexandr Isayev, Alexander Tropsha, Stefano Curtarolo, John Wiley & Sons, 2019

2. Informatics for Materials Science and Engineering-Data-driven Discovery for Accelerated Experimentation and Application, Edited by: Krishna Rajan, Butterworth-Heinemann, 2013

Title	**First-Principles Calculations**	Number	MTL7450
Department	Metallurgical and Materials Engineering	L-T-P [C]	2–0–2 [3]
Offered for	B. Tech., M.Tech.	Type	Program Elective
Pre-requisite			

Objective

1. Introduce different features of electronic and phonon band structure.

Learning Outcomes

1. Understand the working principles of the first-principles calculations.
2. Simulate the electronic and phonon band structure in materials using first-principles calculations.

Contents

Potentials: Construction method, advantages, types of pseudopotentials, Bloch theorem: Expression for Bloch wave, Finite plane-wave basis, energy convergence with the size of basis (5 lectures)
Electronic band structure: Fermi-Dirac distribution, Metals: charge sloshing, numerical issues, smearing methods (5 lectures)
Lattice dynamics: Dynamical properties, harmonic approximation, phonon dispersion curves, force constants, small-displacement methods, density functional perturbation theory, applications: normal mode analysis (8 lectures)
Electron transport: Boltzman transport equation for electrons, constant relaxation time approximation, transport coefficients (4 lectures)
Phonon transport: Boltzman transport equation for phonons, self-consistent phonon lifetimes, mean-free-path spectrum, lattice thermal conductivity (4 lectures)

Laboratory (Computer simulations):
1) Calculation of optical properties.
2) Calculation of magnetic moment in a system.
3) Calculation of the surface energy.
4) Calculation of stacking fault energy.
5) Calculation of stiffness tensor.
6) Performing an ab initio Molecular dynamics simulation.
7) Calculation of the phonon dispersion relation.
8) Calculation of the thermodynamic properties from the phonon dispersion relation.
9) Calculation of Raman spectra
10) Calculation of IR spectra

Text Books

1. Lee, J. G., Computational Materials Science: An Introduction, 2nd edition, CRC Press, 2016
2. Dove, M.T., Introduction to Lattice Dynamics, 1st Edition, Cambridge University Press, 1993
3. Sholl, D. S., and Steckel, J. A., Density Functional Theory: A Practical Introduction, 1st Edition, Wiley, 2009

References

1. Parr, R.G., and Yang, W., Density-Functional Theory of Atoms and Molecules, 1st Edition, Oxford Science Publications, 1994
2. Szabo, A., and Ostlund, N. S., *Modern Quantum Chemistry: Introduction to Advanced Electronic Structure Theory*, Courier Corporation, 1996

Online Course Material

1. Ceder, G. and Marzari, N., 3.320 Atomistic Computer Modeling of Materials (SMA 5107). Spring 2005. Massachusetts Institute of Technology: MIT OpenCourseWare, https://ocw.mit.edu. License: Creative Commons BY-NC-SA

Title	**Advanced Molecular Dynamics**	Number	MTL7XXX
Department	Metallurgical and Materials Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech., M.Tech.	Type	Program Elective
Pre-requisite			

Objectives

1. The aim of this course is to provide the student with an understanding of the various molecular dynamic's methods, their capabilities, and their limitations.

Learning Outcomes

1. Make sound judgements regarding the quality of molecular simulation studies reported in the literature
2. Have exposure to various advanced molecular dynamics techniques to reach experimental level times scale
3. Get an idea to set up the molecular dynamics simulations to solve realistic problems

Contents

Molecular dynamics of hard spheres, demonstrating elementary concepts of temperature, ensemble averaging, error estimation, periodic boundaries, classical potentials and Machine learning potential. Equations of motion, integration schemes, Various ensembles and their applications, dynamical properties-Macroscopic Transport Phenomena (10 Lectures)

Limitations of Equilibrium Methods, Non-Equilibrium Molecular Dynamics, One (Disfavored) Approach, Basic theory and example of static and dynamic linear response, Diffusion: An Inhomogeneous Approach, Self-Diffusion: Perturbation, Thermostatting, Shear Viscosity: Boundary-Driven Algorithm, Limitations of Boundary-Driven Shear (12 Lectures)

Accelerated molecular dynamics, Parallel-Replica Dynamics, Hyperdynamics, Temperature-Accelerated Dynamics, Recent Advances and Applications, Overview of sampling methods (5 Lectures)

Ab-initio molecular dynamics, Basics of Nonadiabatic molecular dynamics, Different flavors of nonadiabatic molecular dynamics (5 Lectures)

Overview of various software to perform molecular dynamics, A model example of molecular dynamics calculation and its result analysis. (7 Lectures)

Books

1. Allen, M.P. and Tildesley, D.J., *Computer Simulation of Liquids*, Clarendon Press, Oxford 1987.
2. Frenkel, D. and Smit, B., *Understanding Molecular Simulation*, second edition, Academic Press, San Diego, 2002.
3. Marx, D. and Hutter, J., *Ab Initio Molecular Dynamics*, Cambridge University Press, Cambridge, 2009.

References:

1. Hiroki Nakamura, *Introduction To Nonadiabatic Dynamics*, World Scientific Press, 2019, https://doi.org/10.1142/11359
2. *Ab Initio Nonadiabatic Quantum Molecular Dynamics*, Chem. Rev. 2018, 118, 7, 3305–3336.
3. *Annual Reports in Computational Chemistry*, Volume 5, 2009, Elsevier B.V., ISSN: 1574-1400, DOI 10.1016/S1574-1400(09)00504-0

Title	**ICME**	Course Number	MTL7xxx
Department	Materials Engineering	L-T-P: C	3-0-0 [3]
Offered for	B. Tech., M.Tech.	Type	Program Elective
Prerequisites	-		

Objectives:

1. To develop an understanding of Integrated Computational Materials Engineering (ICME) principles for linking processing, structure, properties, and performance of materials through multiscale computational modeling.
2. To train students in the use of computational modeling, simulations, and machine learning techniques for materials design, process optimization, and predictive materials engineering.

Learning outcomes:

1. Develop ICME workflows combining simulations, experimental data, and machine learning for the design and optimization of advanced materials and manufacturing processes.
2. Apply computational methods such as CALPHAD, finite element analysis, phase-field modeling, and crystal plasticity to solve materials engineering problems across multiple length scales.

Contents

Module 1: Introduction to ICME (3 Lectures)

Processing-Structure-Property-Performance linkages, need of ICME, multiscale modeling of microstructure evolution, spatial and temporal discretization, numerical solution of partial differential equations, periodic boundary conditions

Module 2: Process modeling (5 lectures)

Finite element modeling of various processes such as forming processes (rolling, deep drawing), joining processes (welding simulations), additive manufacturing, constitutive laws, yield criterions, thermal mechanical coupling, residual stress and distortion prediction.

Module 3: Modeling of microstructure evolution (8 lectures)

Metropolis algorithm, Ising model, Q-states Potts model, 1-D and 2-D cellular automata, applications in microstructure evolution, magnetic spin ordering, isotropic and abnormal grain growth simulations, recrystallization simulations, implementation of experimental grain boundary energy and mobility

Modeling of microstructure deformation, constitutive laws, mean field and full field crystal plasticity models, phenomenological and physical hardening laws, quantification of deformation modes activity, texture evolution, r-value prediction, modeling of dynamic recrystallization.

Module 4: Modeling of diffusion (5 lectures)

Diffusion Mechanisms in Crystalline Solids, Phenomenological Laws of Diffusion, Solutions to Fick's Second Law, Diffusion Forces and Atomic Mobility, A Random Walk Treatment, Application of diffusion modeling, nitriding/carburizing concentration profiles, Diffusion Limited Aggregation.

Module 5: Phase field modeling (5 lectures)

Conserved and non-conserved order parameters, Governing equations, Interface Energy and Diffuse Interface, application to spinodal decomposition in phase separation, application to microstructure evolution in solidification.

Module 6 (13 lectures):

1. FEA simulation of materials under tensile/compression loading.
2. Rolling and deep drawing simulation
3. Mean-field crystal plasticity simulation, R-Value calculation
4. Ising model and Metropolis algorithm for canonical magnetic ordering simulation
5. Q-state model and its application in grain growth simulation
6. 1-D and 2-D cellular automata
7. Nitriding simulation by Fick's second law
8. Dendrite simulation using Diffusion Limited Aggregation
9. Spinodal decomposition microstructure simulation
10. Solidification microstructure simulation by phase field method

Textbooks

1.Janssens, K. G. F., Raabe, D., Kozeschnik, E., Miodownik, M. A., & Nestler, B. (2007). *Computational Materials Engineering: An Introduction to Microstructure Evolution*. Elsevier Academic Press.

2. LeSar, R. (2013). *Introduction to computational materials science: Fundamentals to applications* (1st ed.). Cambridge University Press.

References

1. Janssens, K. G. F., Raabe, D., Kozeschnik, E., Miodownik, M. A., & Nestler, B. (2007). *Computational materials engineering: An introduction to microstructure evolution* (1st ed.). Elsevier Academic Press.
2. Binder, K., & Heermann, D. W. (2019). *Monte Carlo simulation in statistical physics: An introduction* (6th ed.). Springer.

Online Materials

1. Das, P. (2026). *Advanced Manufacturing and Process Modelling* [Online Course]. NPTEL, Indian Institute of Science (IISc), Bengaluru. https://nptel.ac.in/courses/113108631

Title	**Mineral Beneficiation**	Number	MTL 4XXX
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objectives

The Instructor will:

Introduce the principles of physical separation methods, simulation, and instrumentation related to mineral processing

Learning Outcomes

The students are expected to have the ability to:

Ability to engineer selective mineral processing based on material and environment conditions

Contents

Introduction to Mineral Geology [2 lectures]: Geological formation of minerals. Classification and identification of minerals

Introduction to Mineral Beneficiation [4 lectures]: Mineral Beneficiation Plant, Materials handling, Sampling, Sizing of ore particles- sieve analysis and testing, particle size distribution

Comminution and Screening [10 lectures]: Liberalization of dissimilar minerals, Degree of liberalization, comminution- fracture, laws of comminution, crushing – Primary and secondary crusher, crushing operation, Ginding- Ball milling, Wet and Dry Grinding, Chemical Comminution-leaching, dissolution, Screening- types, factors affecting rate of screening, screening efficiency

Separation and Concentration of ores [11 lectures]: Classification mechanical sorting, Gravity separation or concentration- Float and sink, heavy medium separation, Jigging, Flowing film concentration, Froth Floatation- floating reagents, types of floatation, Surface properties and selective flotation of ores, floatation for Sulphide ores, flocculation, Magnetic and Electrical Separation

Solid-Liquid Separation [4 lectures]: Settling of solids in fluids, Solid-Liquid separation-Dewatering, Mechanical Dewatering of Slurries and Liquid-Liquid processing methods, Filtering

Applications and Examples *[8 lectures]:* Mineral beneficiation of Zn, Cu, Fe, Pb ores, Heap leaching of Au, *Mineral Processing of special materials (from brine and mineral resources), rare earths, vanadium, etc.*

Textbooks

1. Wills,B.A. and Napier-Munn,T.J., (2006), Mineral *Processing Technology*, Elsevier Science & Technology Books, 7th Edition
2. Subbarao,D.V., (2011), *Mineral Beneficiation: A Concise Basic Course*, CRC Press

Reference Books

1. Drzymala, J., (2007), Mineral Processing: foundations of theory and practice of minerallurgy, Wroclaw University of Technology
2. German,R.M., (2005), *Powder Metallurgy & Particulate Materials Processing*, Metal Powder Industry

Online Course Materials

1. Majmuder, A., *Introduction to Mineral Processing*, NPTEL Course Material, Department of Mining Engineering, Indian Institute of Technology Kharagpur, http://nptel.ac.in/courses/105105171/

Title	**Extraction of Nonferrous Metals**	Number	MTL 4XXX
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objectives

Explain principles of thermodynamics and process metallurgy involved in traditional and advanced processes of nonferrous metal extraction

Learning Outcomes

1. Understand the basics of extractive metallurgy involved in nonferrous metal extraction processes

Contents

Early developments in metal extraction [2 lectures]: Nonferrous metals in Indian history, uses of nonferrous metals, Sources of nonferrous metals

Principles of metals extraction [4 lectures]: Thermodynamic principles, homogeneous and heterogeneous reactions, Ellingham diagram, kinetic principles, principles of electro-chemistry

General methods of extraction [7 lectures]: Pyrometallurgy – calcination, roasting and smelting, Hydrometallurgy – leaching, solvent extraction, ion exchange, precipitation, and electrometallurgy – electrolysis and electro- refining)

General methods of refining [4 lectures]: Basic approaches, preparation of pure compounds, purification of crude metal produced in bulk

Extraction of metals from oxide sources [6 lectures]: Basic approaches and special features of specific extraction processes, extraction of metals such as magnesium, aluminum, tin and ferro-alloying elements, production of ferro alloys.

Extraction of metals from Sulphide Ores [6 lectures]: Pyro-metallurgy and hydrometallurgy of sulphides, production of metals such as copper, lead, zinc, nickel etc.

Extraction of metals from Halides [6 lectures]: Extraction of metals from halides: Production of halides and refining methods, Methods of extraction of metals such as titanium, rare earths, uranium, thorium, plutonium, beryllium, zirconium etc.

Production of precious metals [2 lectures]: Methods applied for gold, silver and Pt group of metals

Waste Utilization [2 lectures]: Secondary metals and utilization of wastes, Energy and environmental issues in nonferrous metals extraction

Text Books

1. Ray, H.S., Sridhar, R. and Abraham, K.P. Extraction of nonferrous metals, Affiliated East West Press Pvt Ltd., New Delhi (2007)
2. Dennis, W.H., Extractive Metallurgy, Philosophical Library, New York (1965)

Self-Learning Materials

1. Ray, H.S., and Ghosh, A., Principles of extractive metallurgy, Wiley Eastern Ltd., New Delhi, 1991

Online Course Material

1. Ray, H.S., Non-ferrous Extractive Metallurgy, NPTEL Course Material, Department of Materials and Metallurgical Engineering, Indian Institute of Technology Kharagpur, https://nptel.ac.in/courses/113/105/113105021/

Title	**Electrometallurgy**	Number	MTL 4XXX
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objectives

The aims to cover

1. The principles and applications of aqueous electrochemistry in extractive metallurgy.
2. Analysis of the thermodynamic and electro-chemical kinetic phenomena in aqueous and non-aqueous electrowinning and electrorefining of metals.

Learning Outcomes

The student should be able to:

1. Have knowledge and understanding of the existing electrometallurgical processes,
2. Analyze, conceive and conceptually design novel electrometallurgical processes

Contents

Introduction, Definitions, Principles and Concepts [8 lectures]: Electrodes and electrochemical reactions, cell and circuit, The Nernst equation and energy producing cells, Electrolysis, Faraday's law, Electro-crystallization.

Surface Morphology of Metal Electrodeposits [10 lectures]: Thin Compact Surface Metal Films, Thick Compact Metal Electrodeposits, Nucleation rate and deposition overpotential, Deposition from simple/ complex salt solutions with/without the presence of adsorbed additives, Disperse (Spongy, Dendritic, Powdered, Granular, Whisker) Deposits.

The Current Distribution in Electrochemical Cells [10 lectures]: Two equal plane parallel electrodes arrangement, Ohmic resistance of the cell, The edge effect, Cell voltage-current density dependencies, Determination of the current density distribution, Cells with Low Anode Polarisation, Corner Weakness Phenomena in Electroforming, Electrodeposition at a periodically changing rate.

Applications [11 lectures]: Electrowinning (zinc and aluminium), Electrorefining (copper), Optimum conditions for electroplating (metals/alloys/composites) from aqueous/non-aqueous/molten salt solutions, Electropolishing, Electro-machining, Electrochemical Oxidation of Metals.

Textbook

1. Popov K. I., Djokic S. S. and Grgur B. N. (2002) *Fundamental aspects of electrometallurgy*, Kluwer academic publishers, New York.

Reference Books

1. Free M., Moats M., Houlachi G., Asselin E., Allanore A., Yurko J., Wang S (2012) *Electrometallurgy 2012*, John Wiley & Sons, Inc., Hoboken, New Jersey.

Online Course Material

1. Prof. H.S. Ray, Non-ferrous Extractive Metallurgy, NPTEL Course Material, Metallurgy & Material Science, Indian Institute of Technology Kharagpur, https://nptel.ac.in/courses/113105021/

Title	**Iron and Steel Making**	Number	MTL 3080
Department	Materials Engineering	L – T – P [C]	3 – 0 – 0 [3]
Offered for	B. Tech. (MT)	Type	Program Elective (PE)
Pre-requisite			

Objectives

1. Explain principles of thermodynamics and process metallurgy involved in traditional and advanced iron and steel making processes

Learning Outcomes

1. Understand the basics of process metallurgy involved in iron and steel making processes
2. Be cognizant of advanced iron making and steel making processes in industries

Contents

Introduction to Iron Making and steelmaking [10 lectures]: concept of integrated steel plant, different types of ironmaking processes, agglomeration techniques, basic thermodynamics, kinetics and transport phenomena in iron and steelmaking processes

Transport Processes in Blast Furnace [8 lectures]: temperature, burden distribution, and aerodynamics of blast furnace, physical and chemical; processes in upper and lower zones of blast furnace

Alternative Iron Making Processes [8 lectures]: direct reduction process-DRI, HBI, smelting reduction process-Corex, Romelt, HiSmelt, Finex etc.

Modern Steelmaking Processes [7 lectures]: basic oxygen furnace steelmaking, Electric Arc Furnace, and induction furnace steelmaking

Secondary Steelmaking Processes [6 lectures]: deoxidation, ladle and tundish Metallurgy, continuous casting, casting defects

Text Books

1. Ghosh,A., and Chatterji,A., (2008), Ironmaking and Steelmaking: Theory and Practice, Prentice-Hall (India)
2. Turkdogan,E.T., (1997), A Text Book of Steelmaking, Academic Press, London Shetty, M. N., (2013), Dislocations and Mechanical Behavior of Materials, Prentice Hall India Learning Private Limited
3. Chatterjee,A., (1994), Beyond the Blast Furnace, CRC Press

Self-Learning Materials

1. Ghosh,A., (2000), Secondary Steelmaking, CRC Press, Boca Raton Online Course Material
2. Muzumdar,D., and Koria,S.C., Steelmaking, NPTEL Course Material, Department of Material Science and Engineering, Indian Institute of Technology Kanpur, http://nptel.ac.in/courses/113104013/
3. Gupta,G.S., Iron Making, NPTEL Course Material, Department of Materials Engineering, Indian Institute of Science Bangalore, http://nptel.ac.in/courses/113108079/